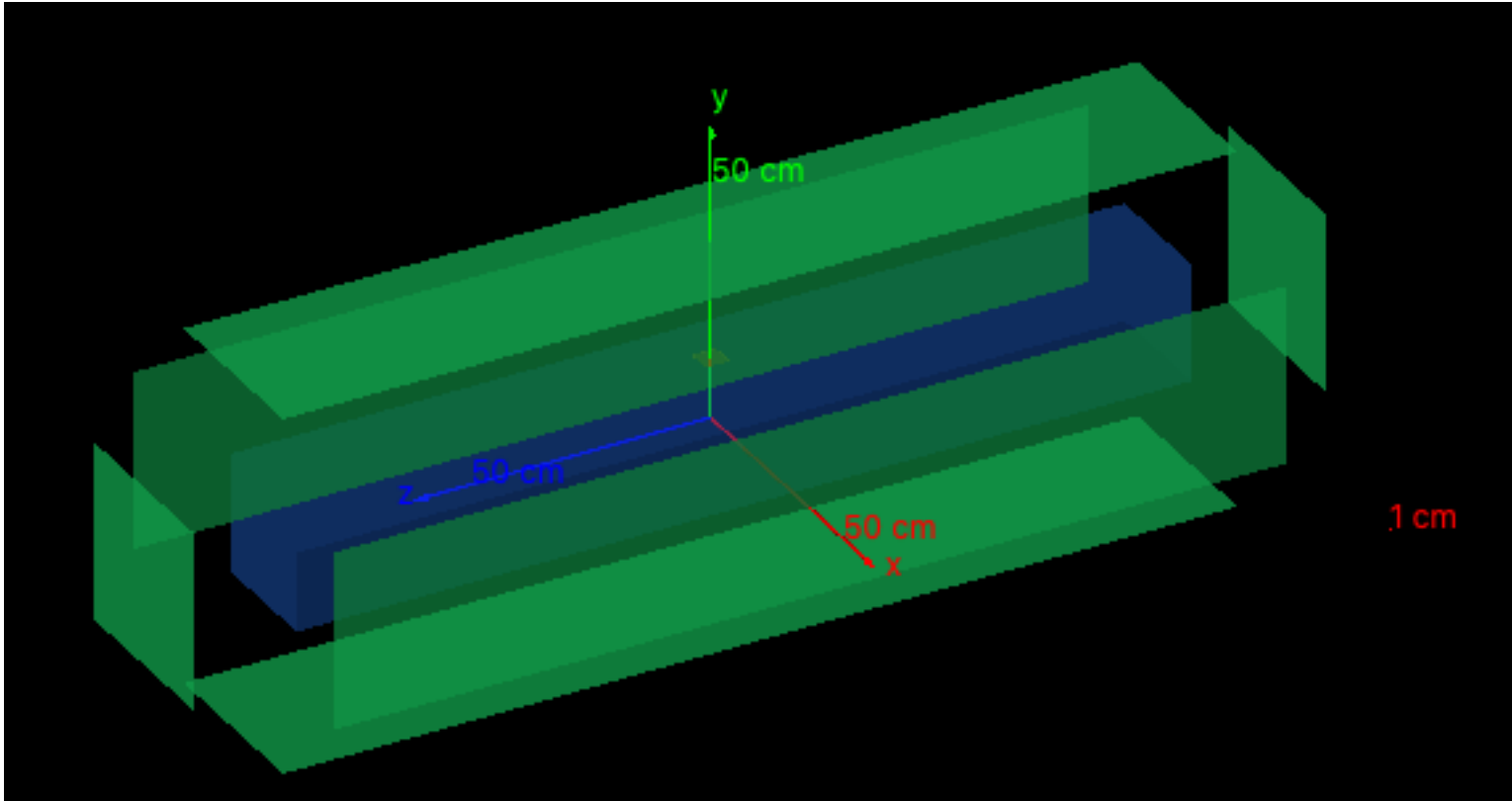


Theranostics playground

Chiara

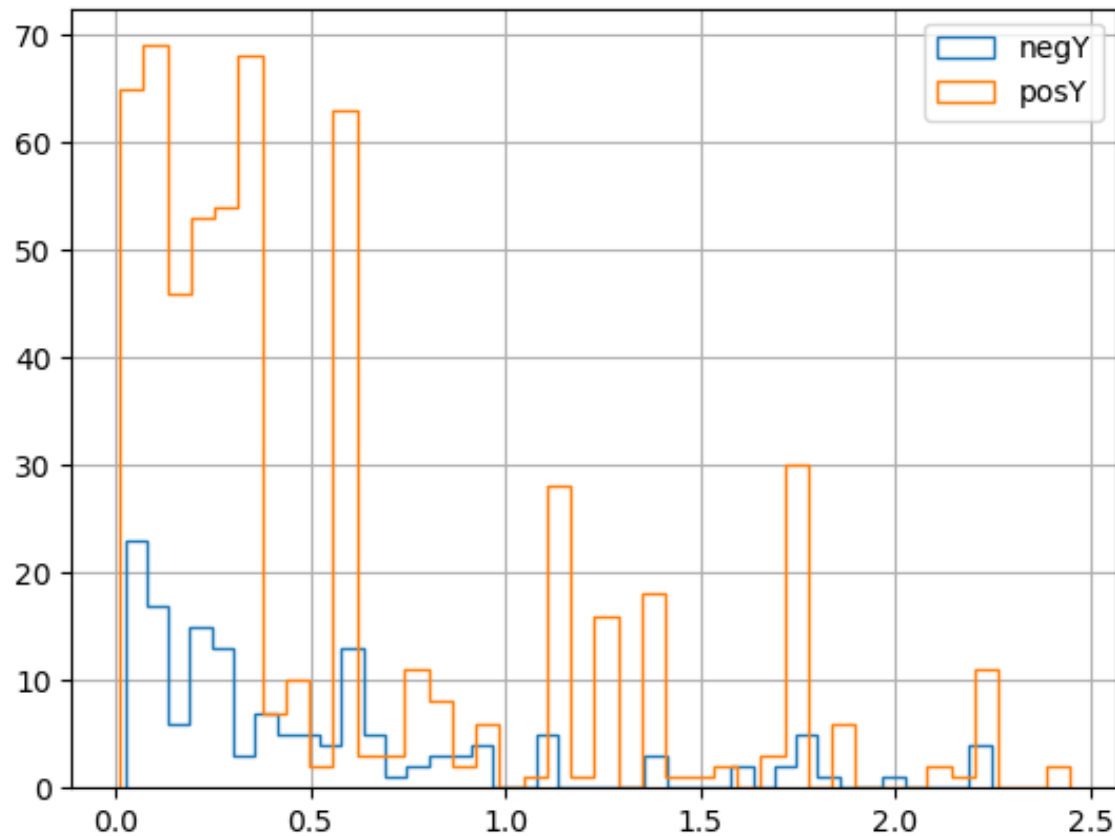
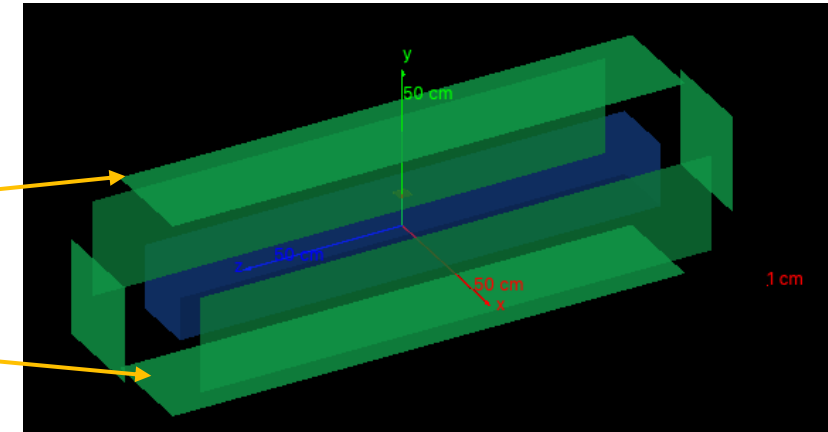
Geant4 Simulation



- 20x20x150 cm water box: body
- 20x20 (+10cm padding) in z
- 20x150 (+10 cm padding) in X and Y
- 4x4 cm glue blob, 2mm thick, with variable loaded portion, on top of the water box, in the centre
- Collecting all photons crossing boundaries and reaching panels, saving tracking information
- Quick runs: 1000 events, fully loaded 2mm, and 50um top-loaded 2mm glue

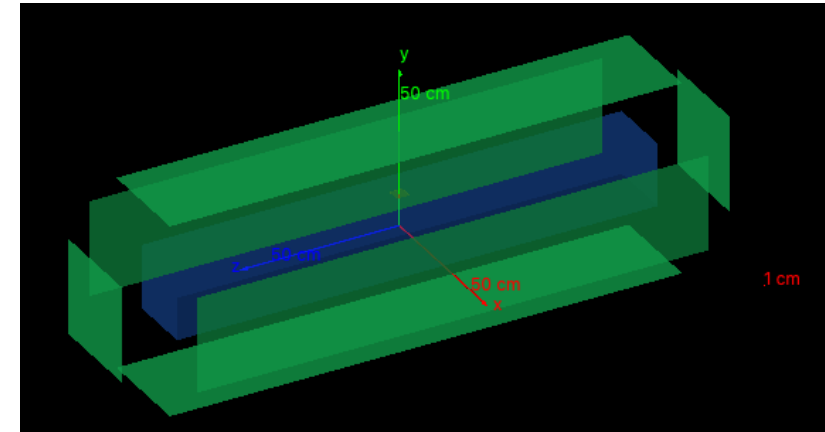
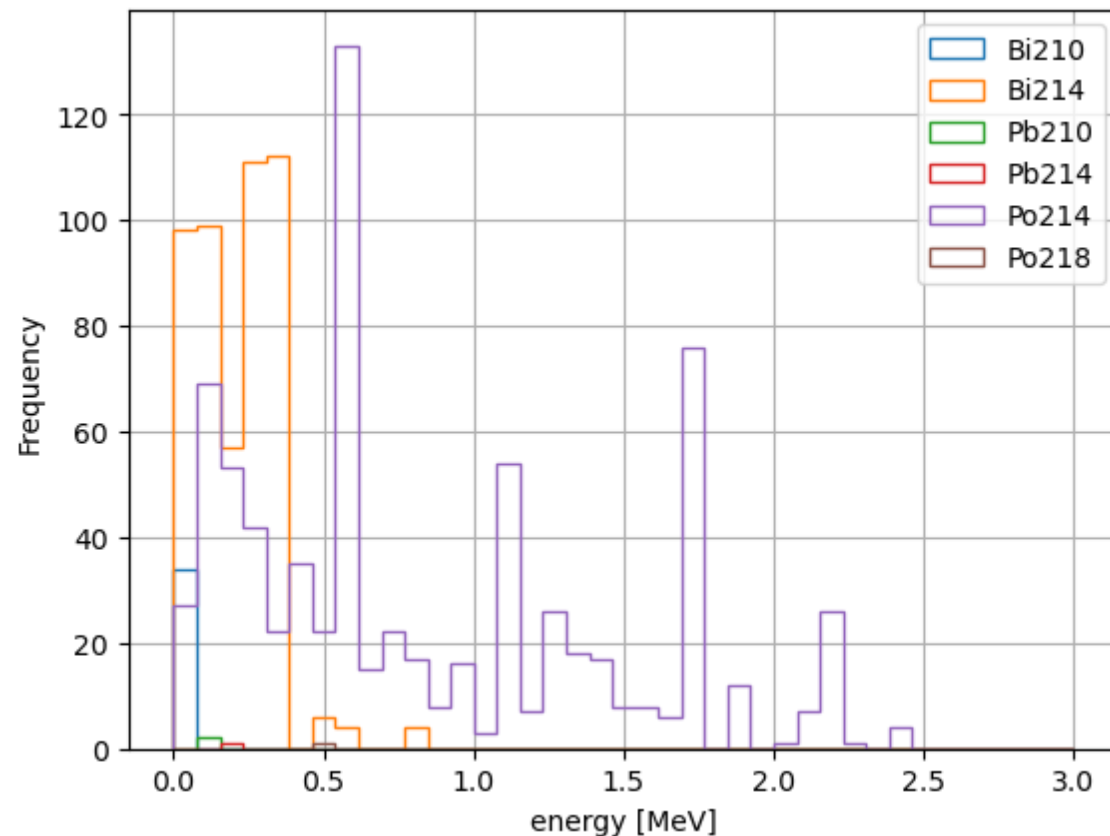
Energy distribution

- Photons entering Y panels (top and bottom)



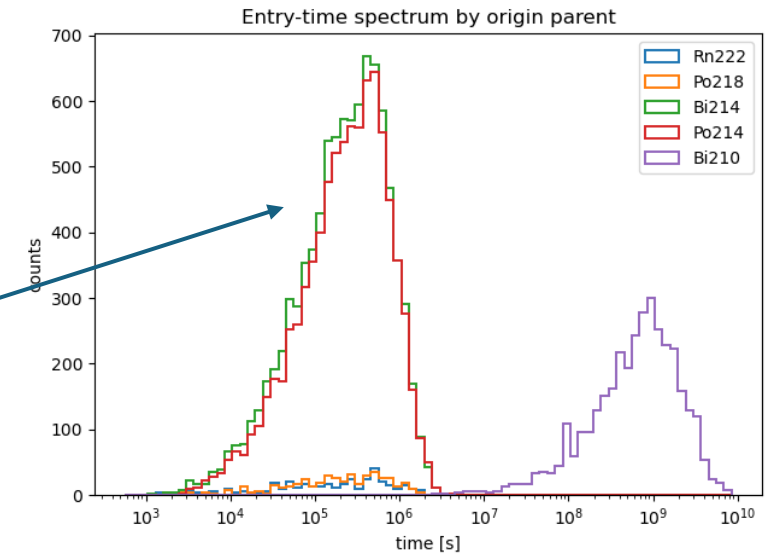
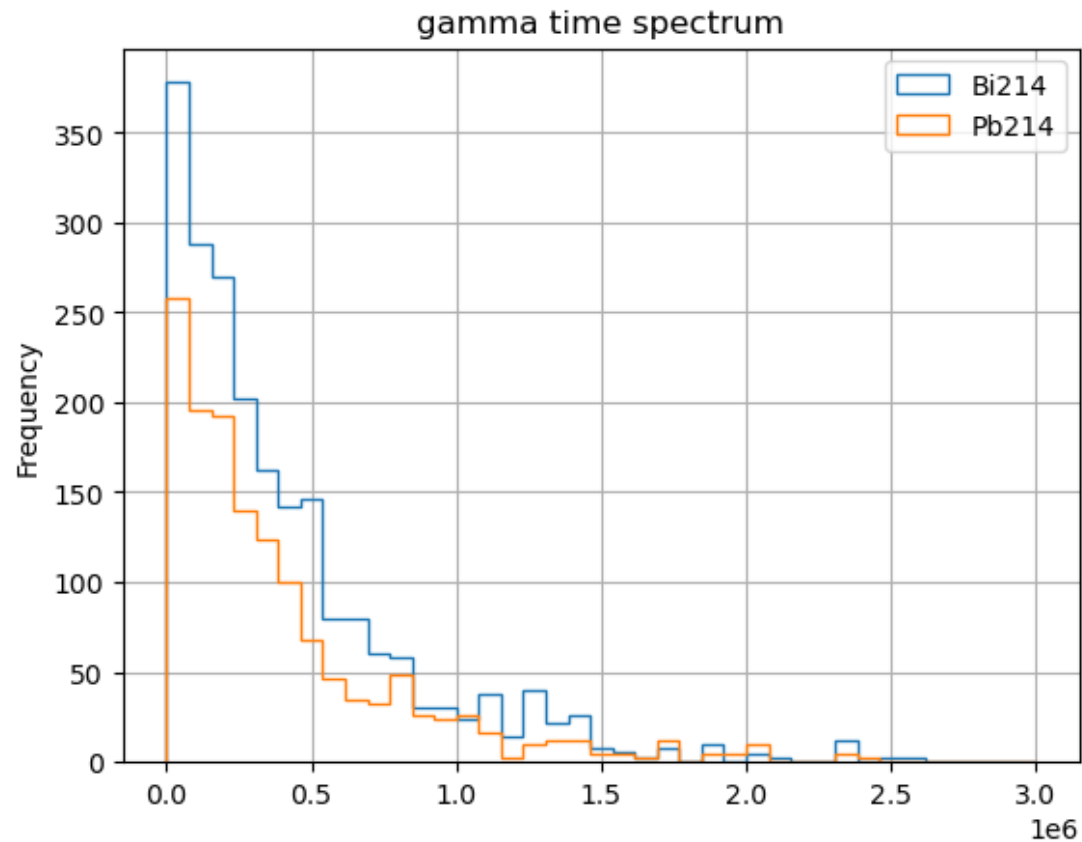
Energy distributions

- Photons entering any panel, aggregated by emitter



Time spectrum

- Bi214 and Pb 214 are in the first peak



Detected gammas

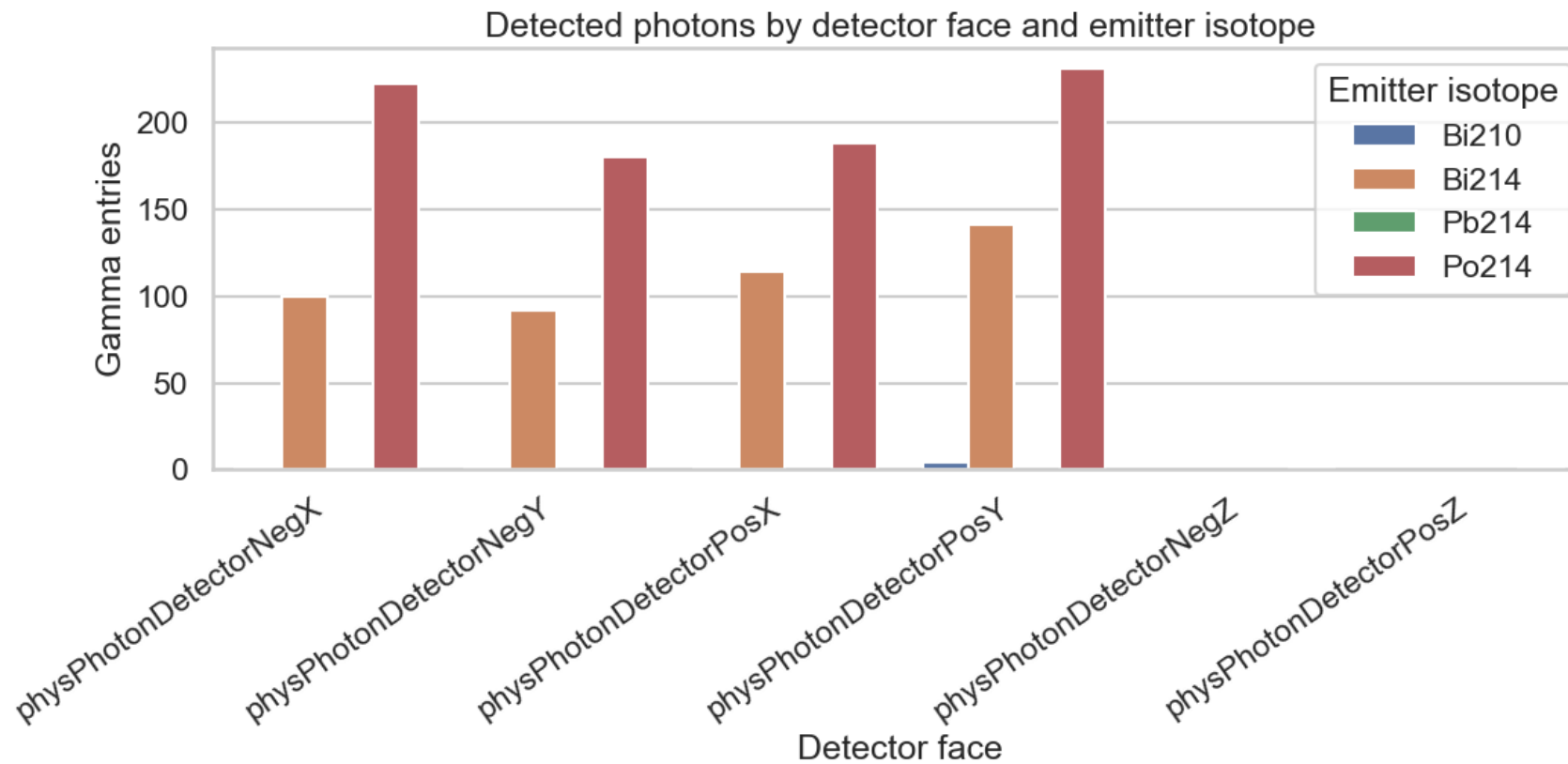
- Detector efficiency (detected/emitted gammas)



	emitted	detected	efficiency
originIsotope			
Pb214	1	1	1.000000
Po214	833	820	0.984394
Bi214	454	444	0.977974
Bi210	16	10	0.625000

Detected gammas

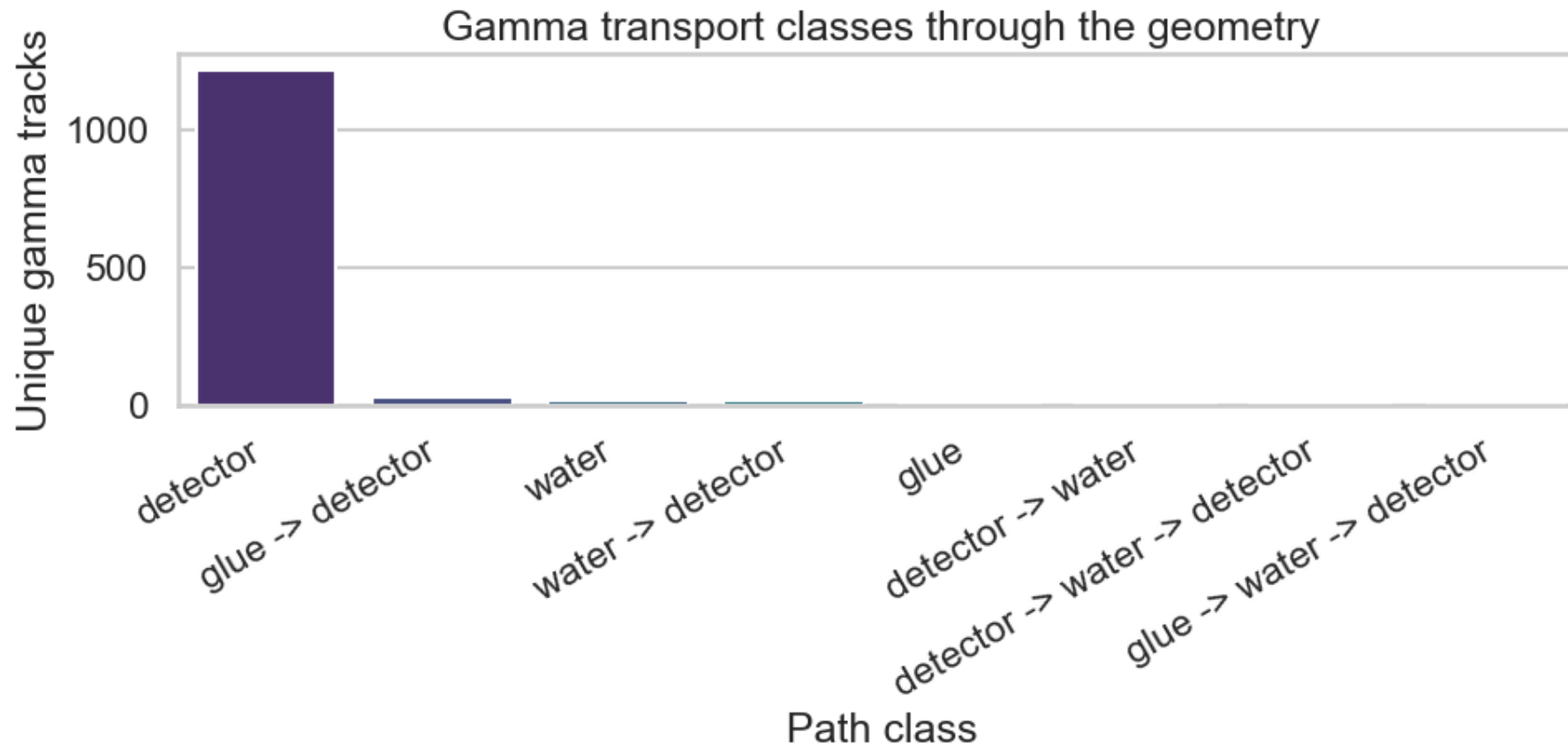
- By panel and emitter isotope



Nothing detected in Z panels – too far away.

Detected gammas

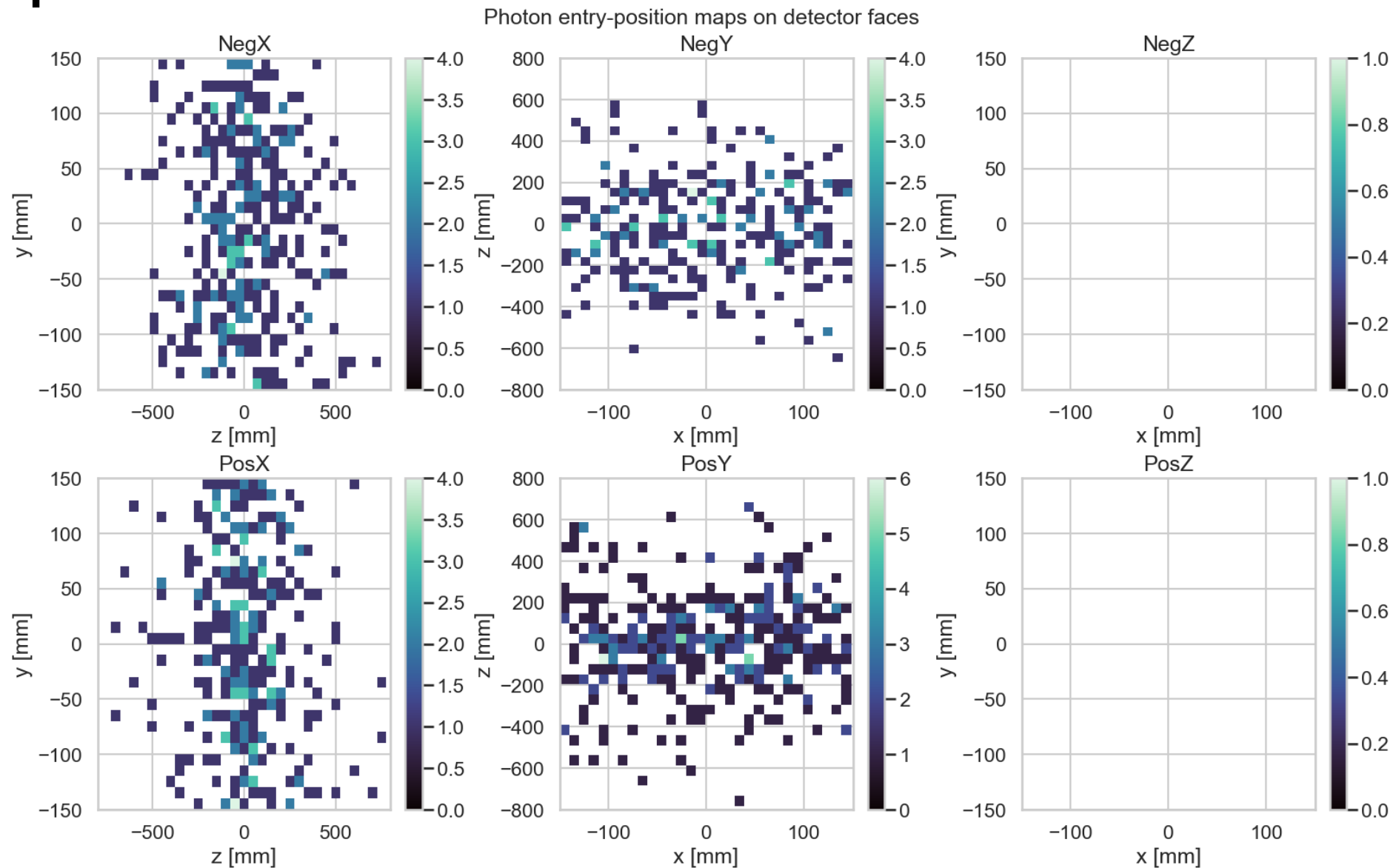
- Trying to understand their path



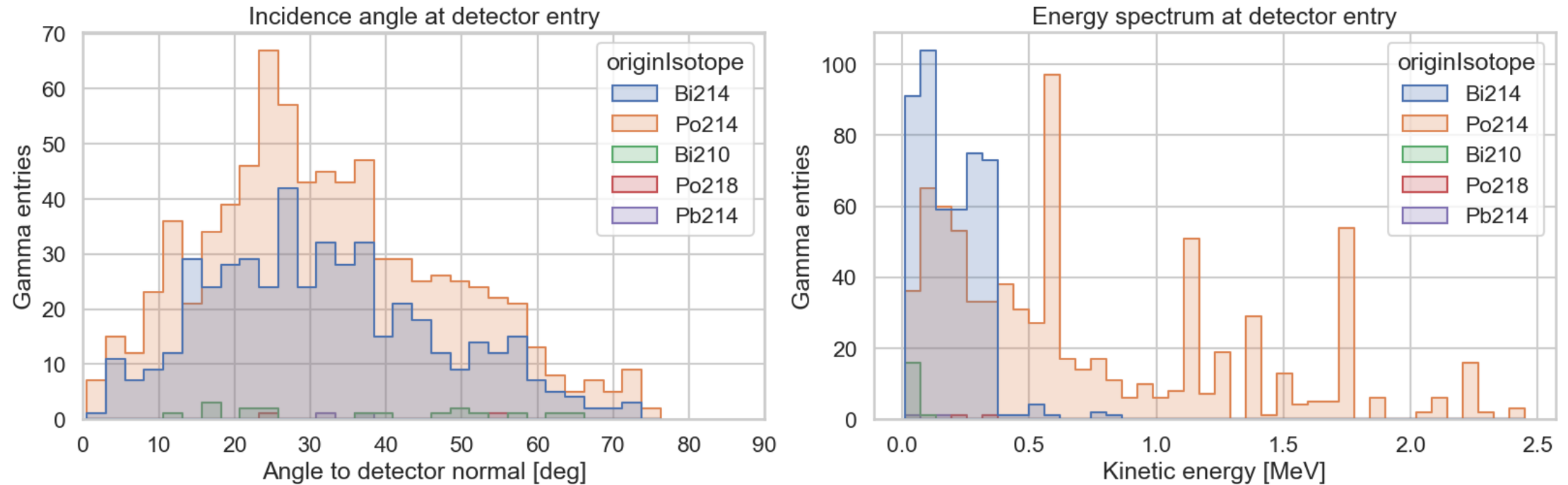
Need to look into this.
Most gammas
produced in the world
volume??

“Empty” space
between water box
and detector

Hit maps



Incident angle and energy spectrum

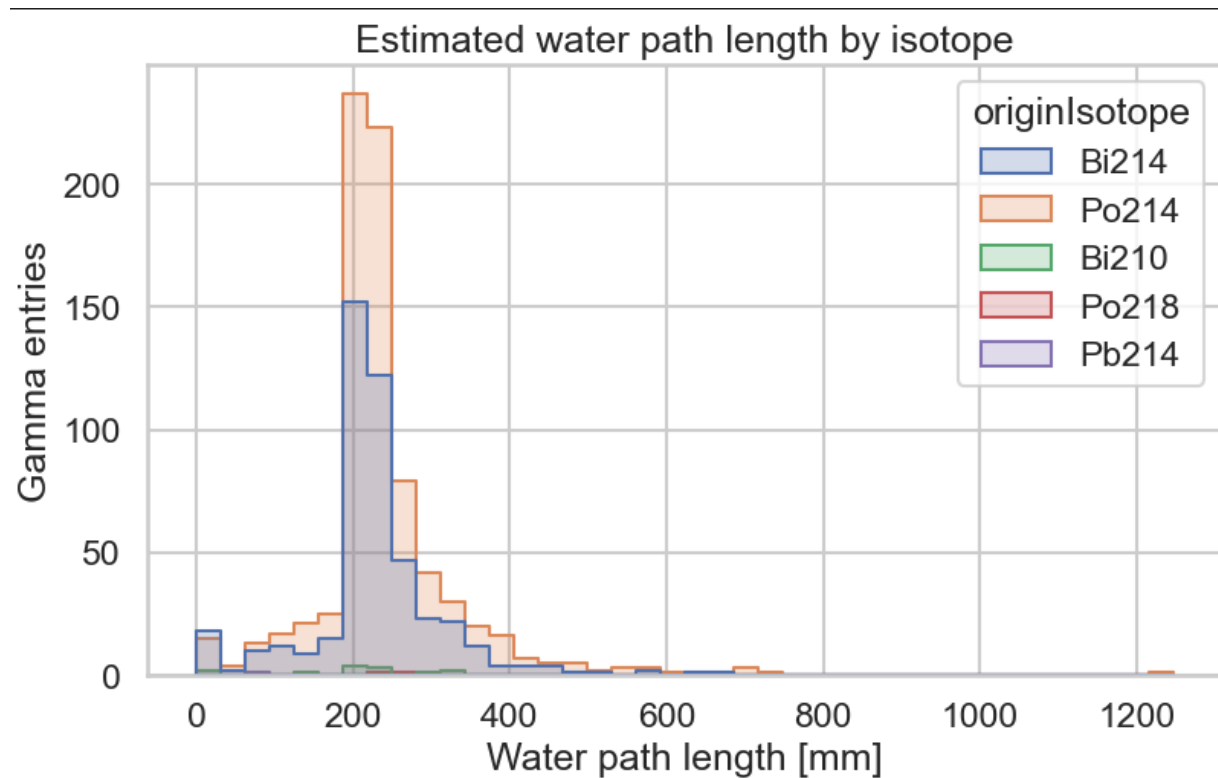


Attempting track reconstruction

- Using detector incident position and momentum to back-project the gamma track into the water box
- Defining a preliminary source-position from the midpoint of the estimated track segment in water

Attempting track reconstruction

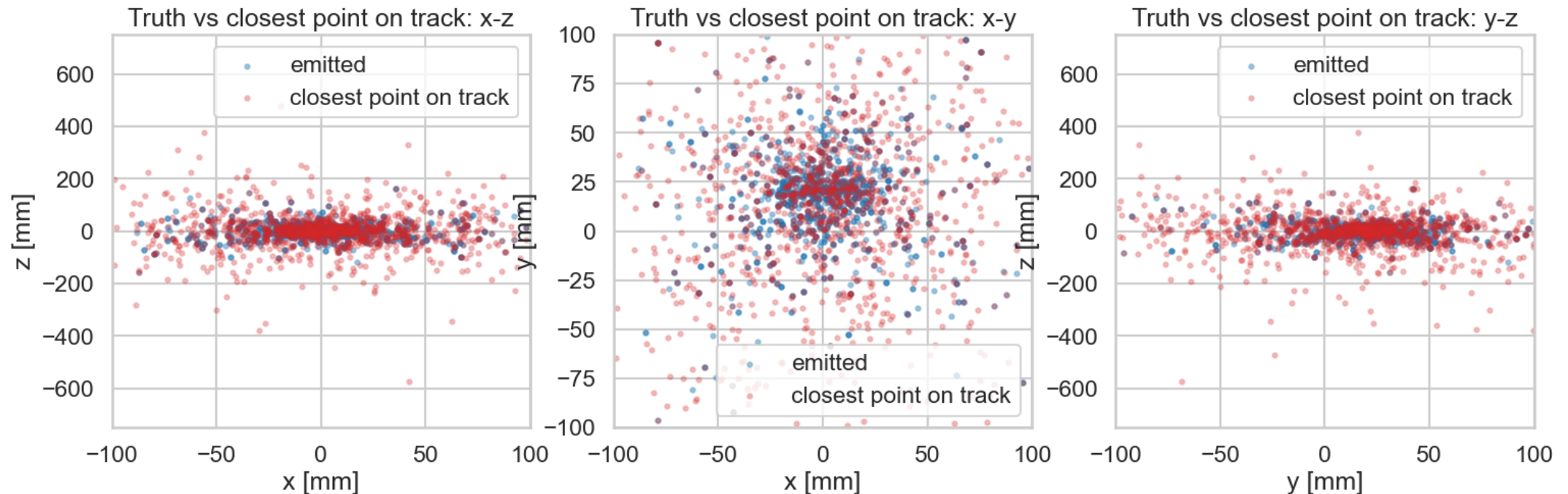
- Estimated path length by isotope (length of the estimated track segment inside the water box)



1. Logged water boundary available:
 - using first and last recorded water boundary points for each gamma track
2. No logged water boundary:
 - starting from hit position, using direction from detected momentum, Intersect with water box: get 2 points - > length

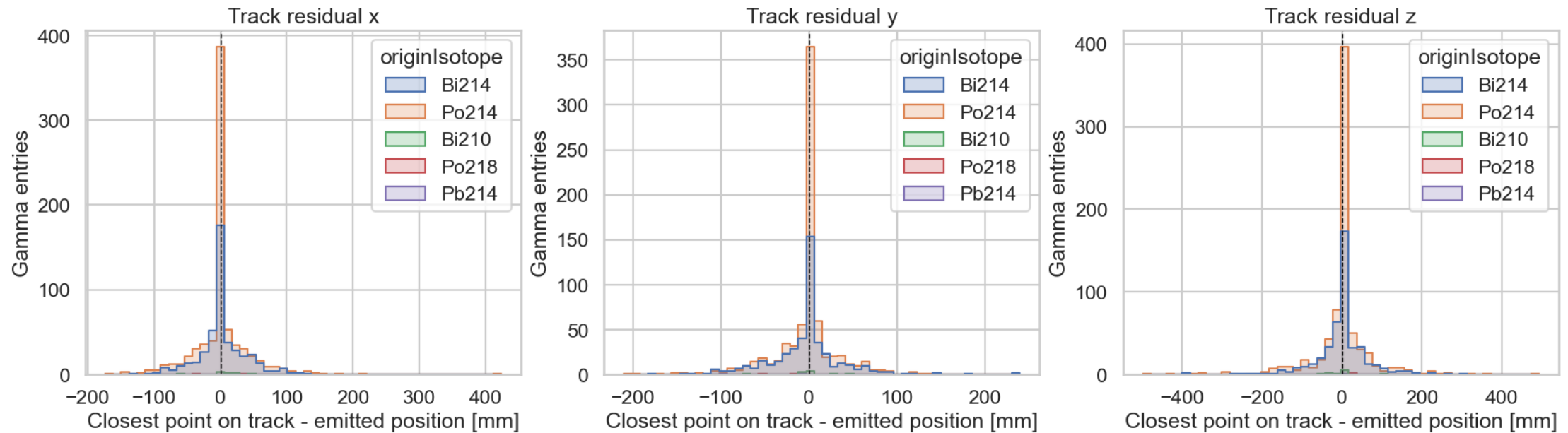
Attempting track reconstruction

- Comparing estimated source position to “real” simulated first position of gamma - preliminary



Attempting track reconstruction

- Comparing estimated source position to “real” simulated first position of gamma - preliminary



Detecting diffusing sources

Default 2mm glue in the centre of the “body”, plus different options:

- Part of the events with Pb214 diffusing in the whole body (isotopes generated in the whole volume)
- Part of the events with Pb214 in the whole body, and Bi214 in a localized area (5x5x5cm) displaced by 50cm

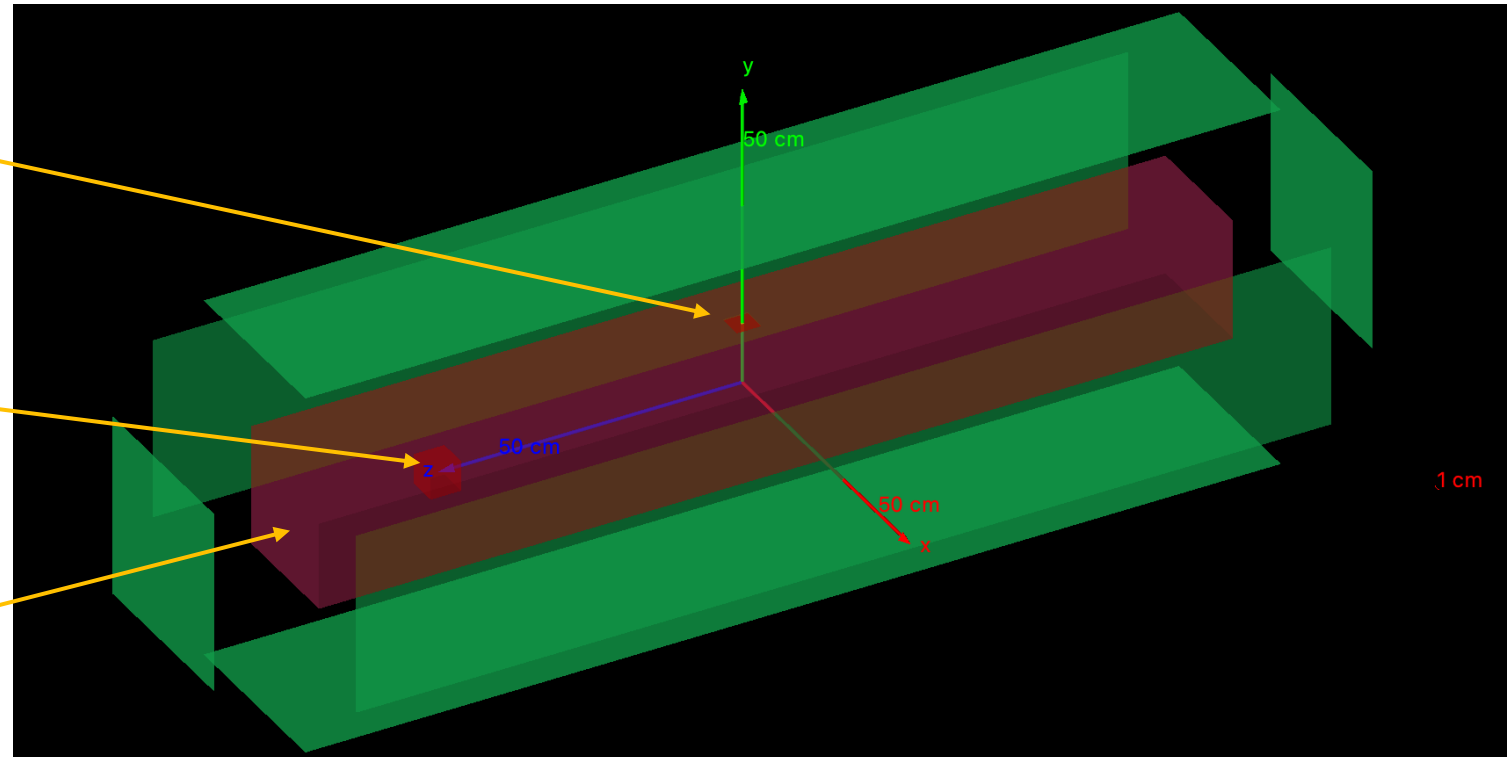
Detecting diffusing sources

- Visualising different GPSs

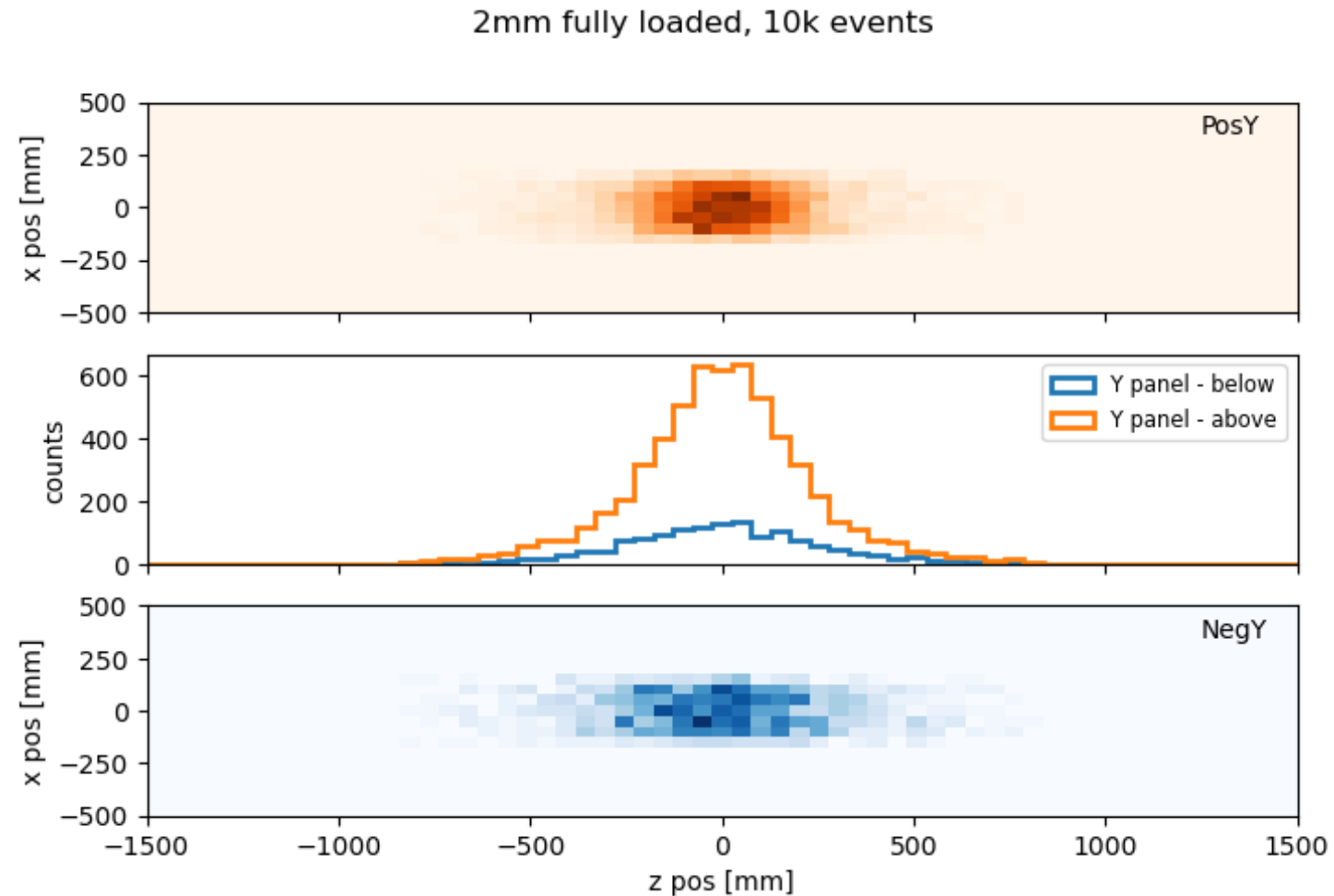
Rn222

Bi214

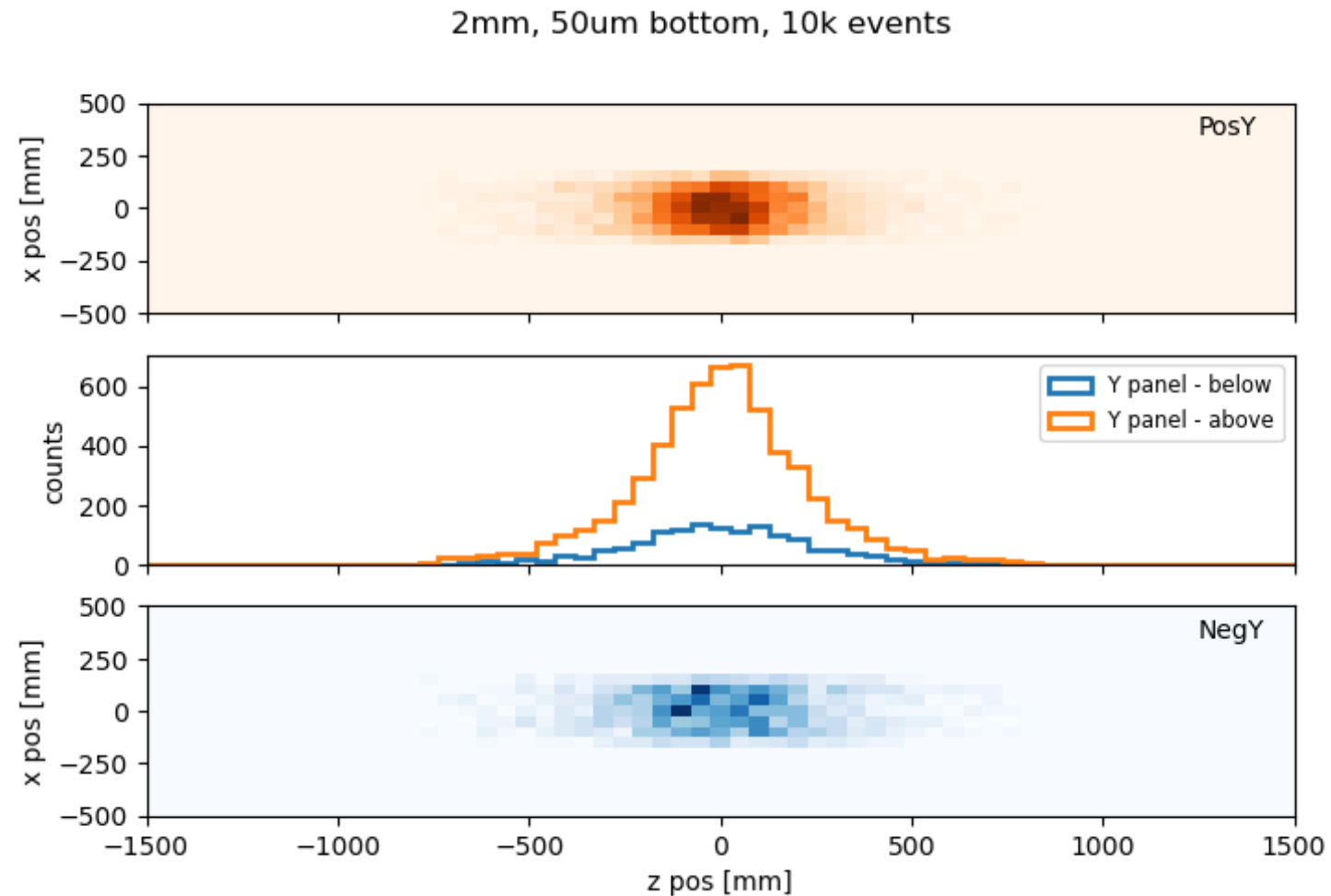
Pb214



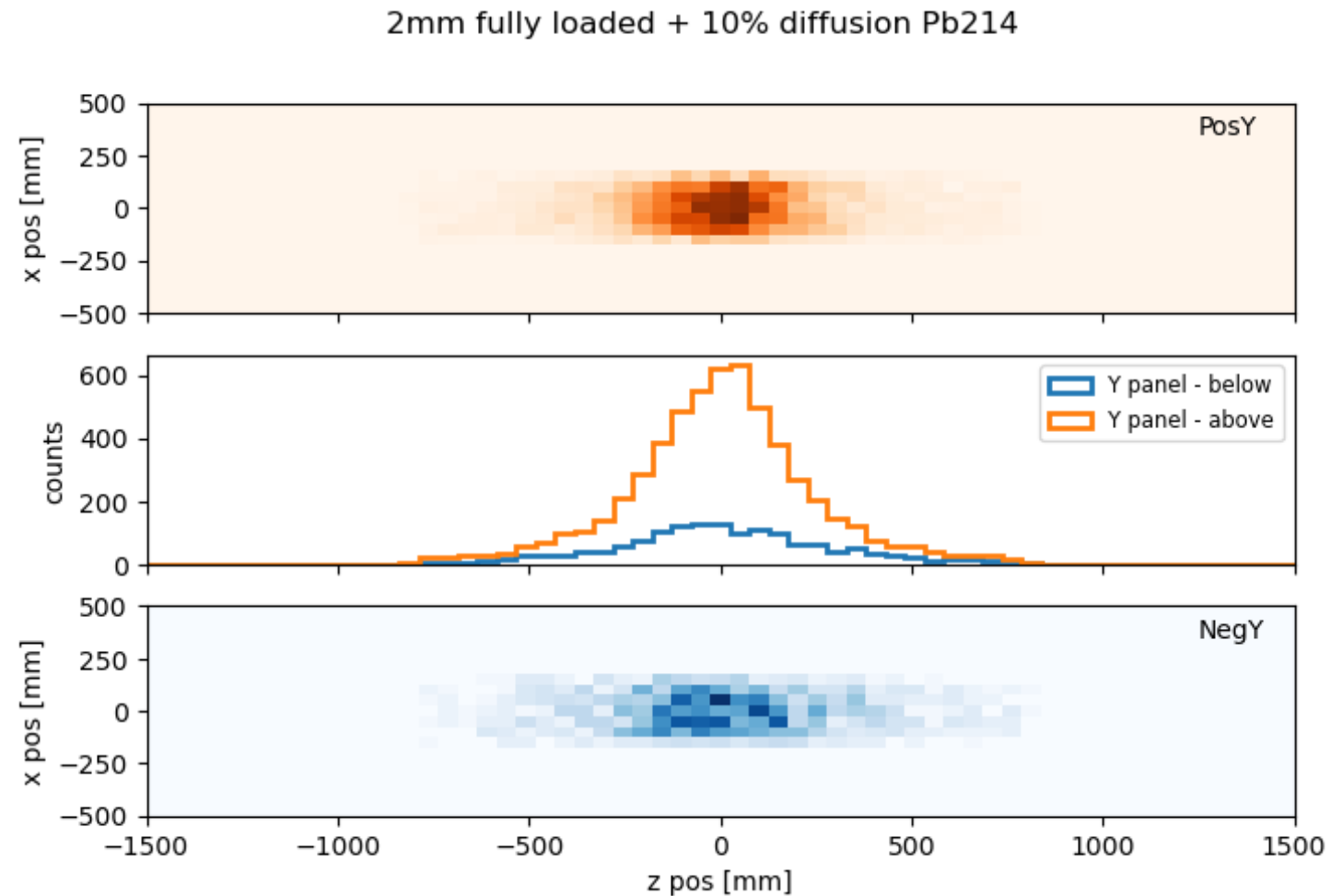
Fully loaded glue – default case



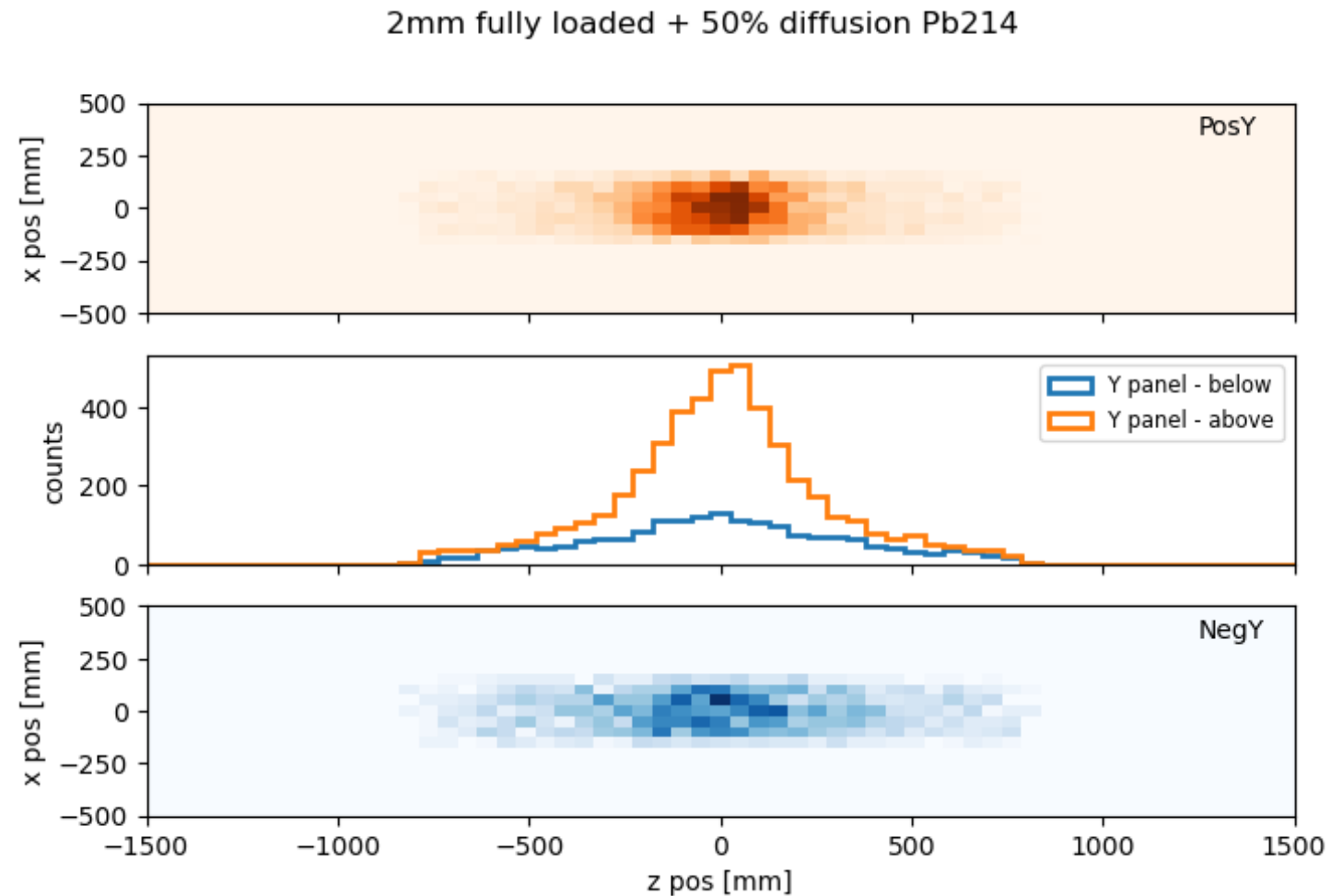
Fully loaded glue – default case



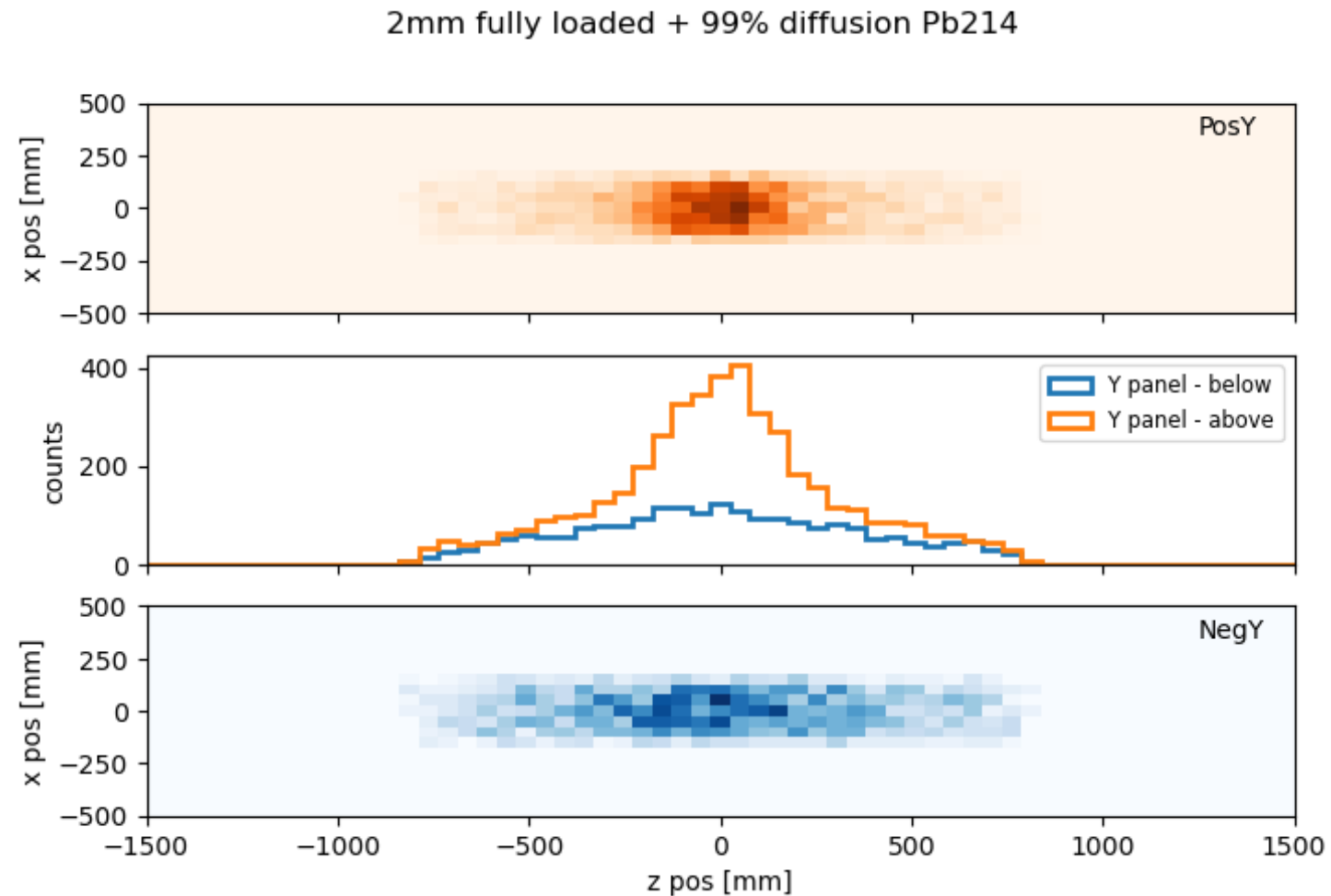
Adding diffusion: 10% Pb214 whole body



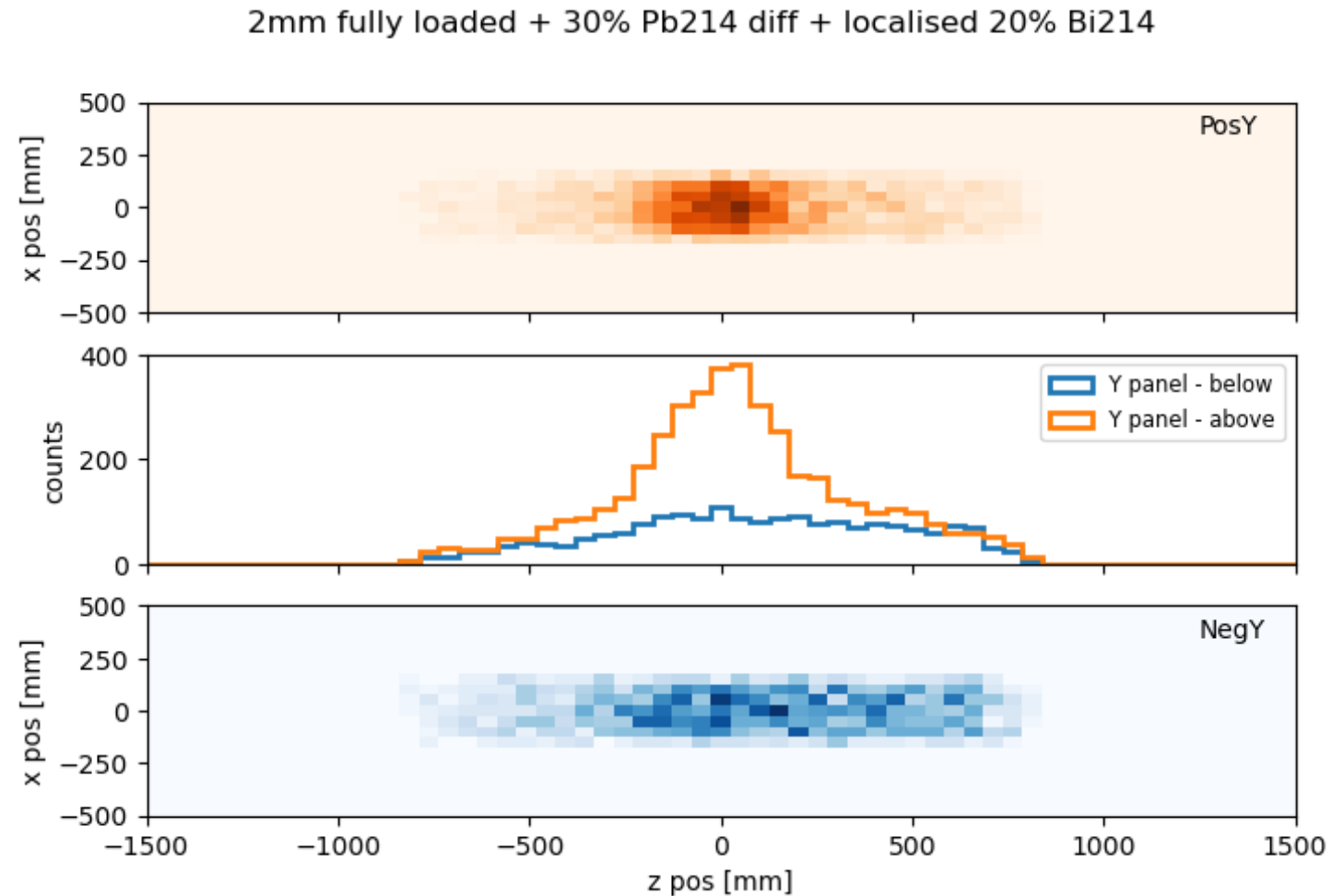
Adding diffusion: 50% Pb214 whole body



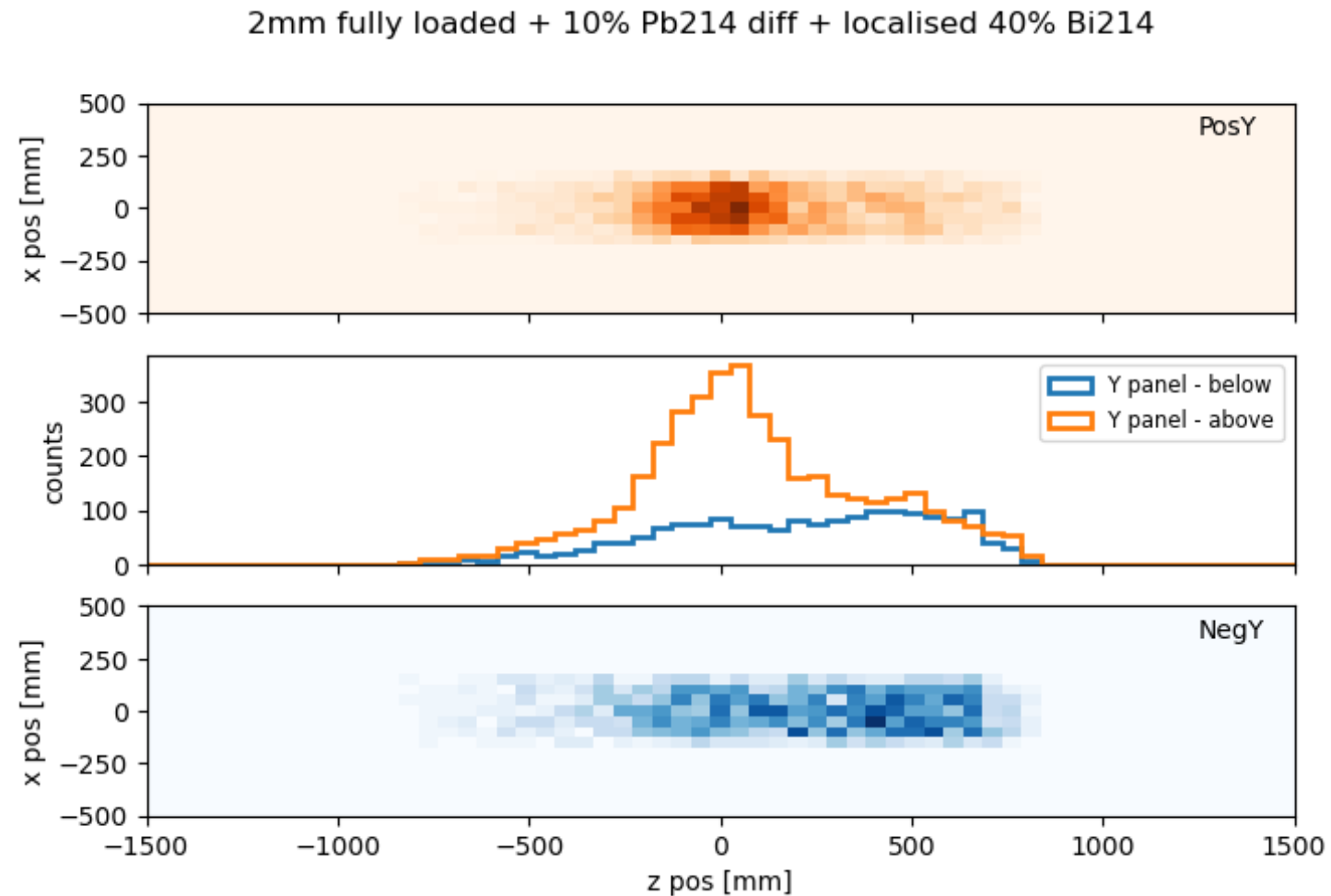
Adding diffusion: 99% Pb214 whole body



Adding diffusion 30% Pb214 + 20% Bi214 localised

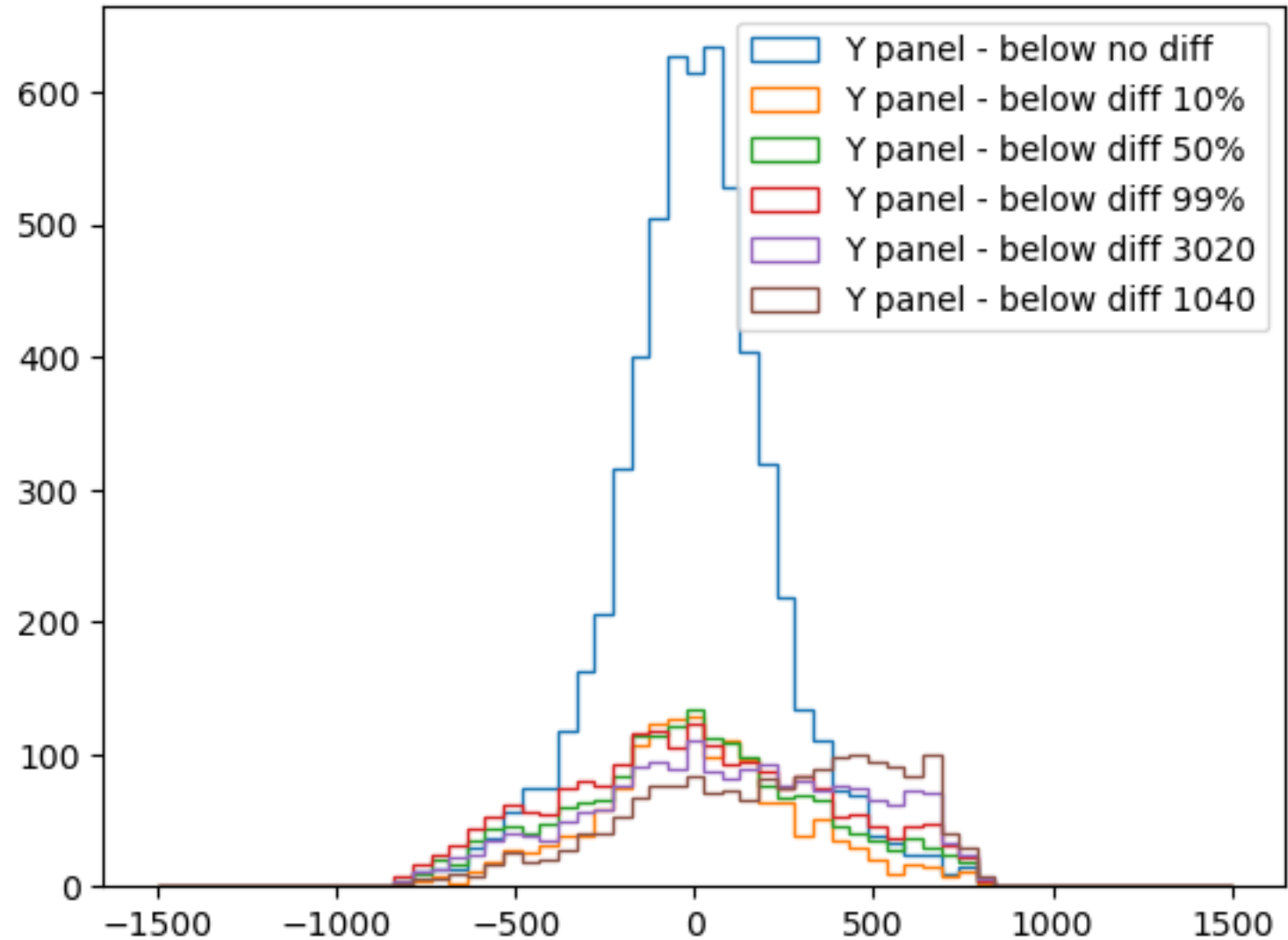


Adding diffusion 30% Pb214 + 20% Bi214 localised



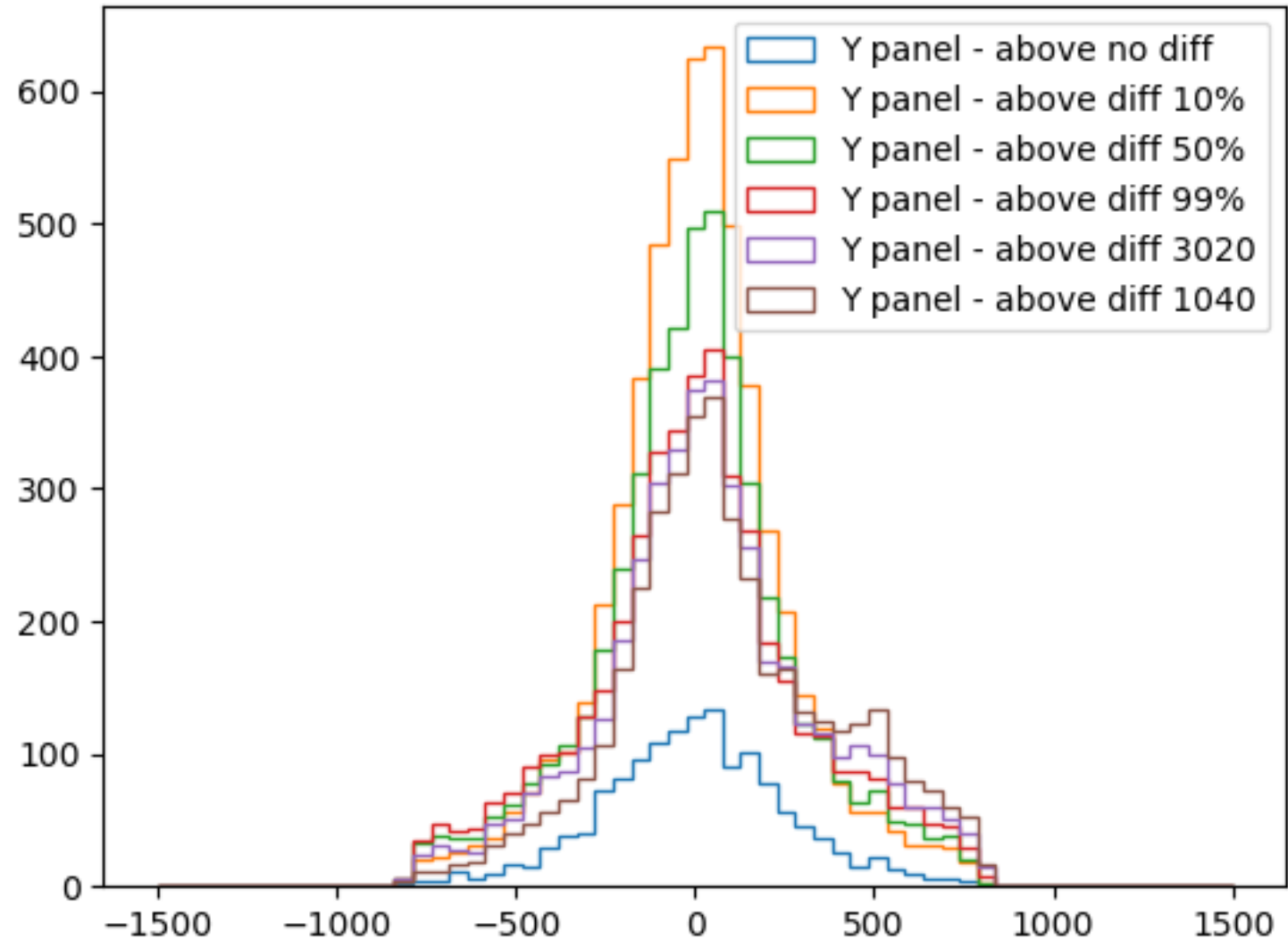
Profile

- Z position in Y panels
- Panel below body



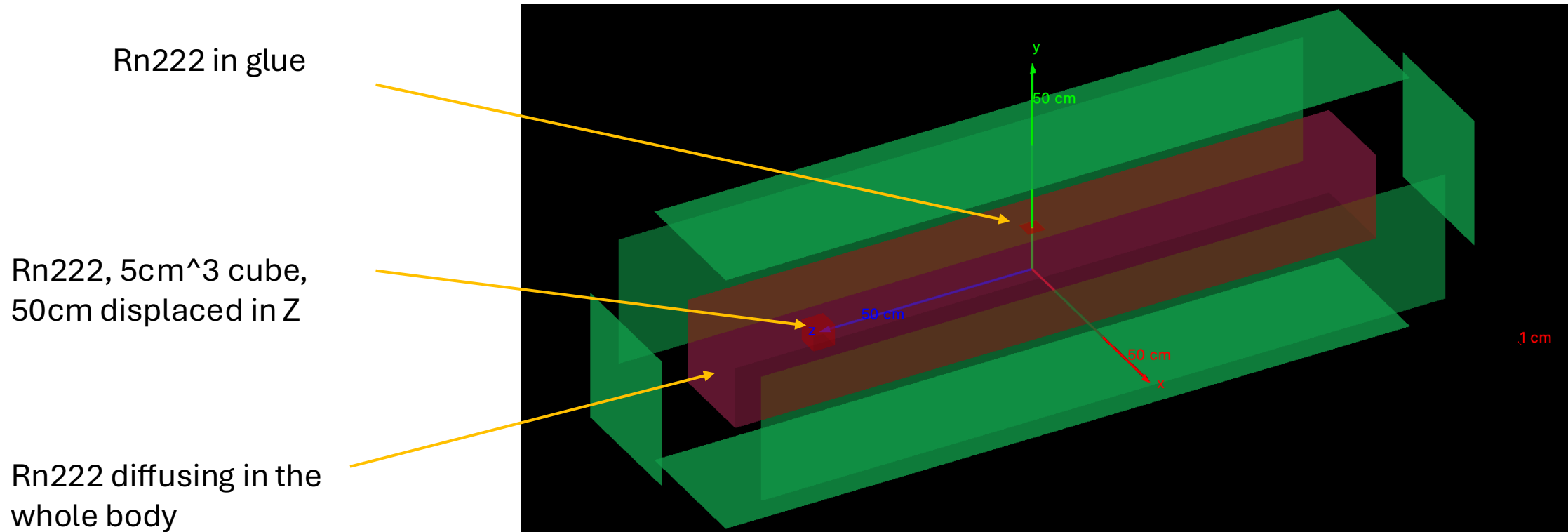
Profile

- Z position in Y panels
- Panel above body

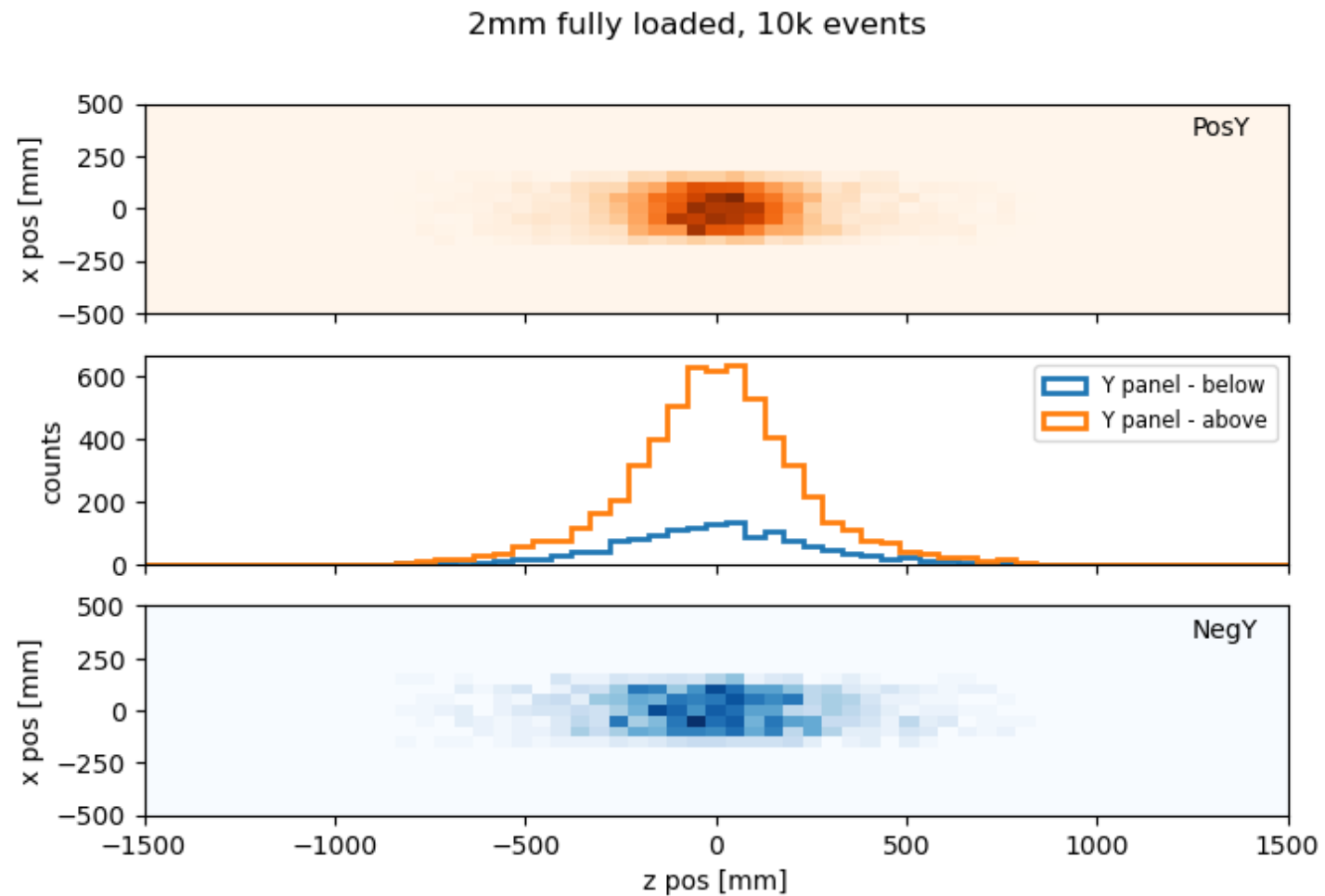


Rn222 diffusion only

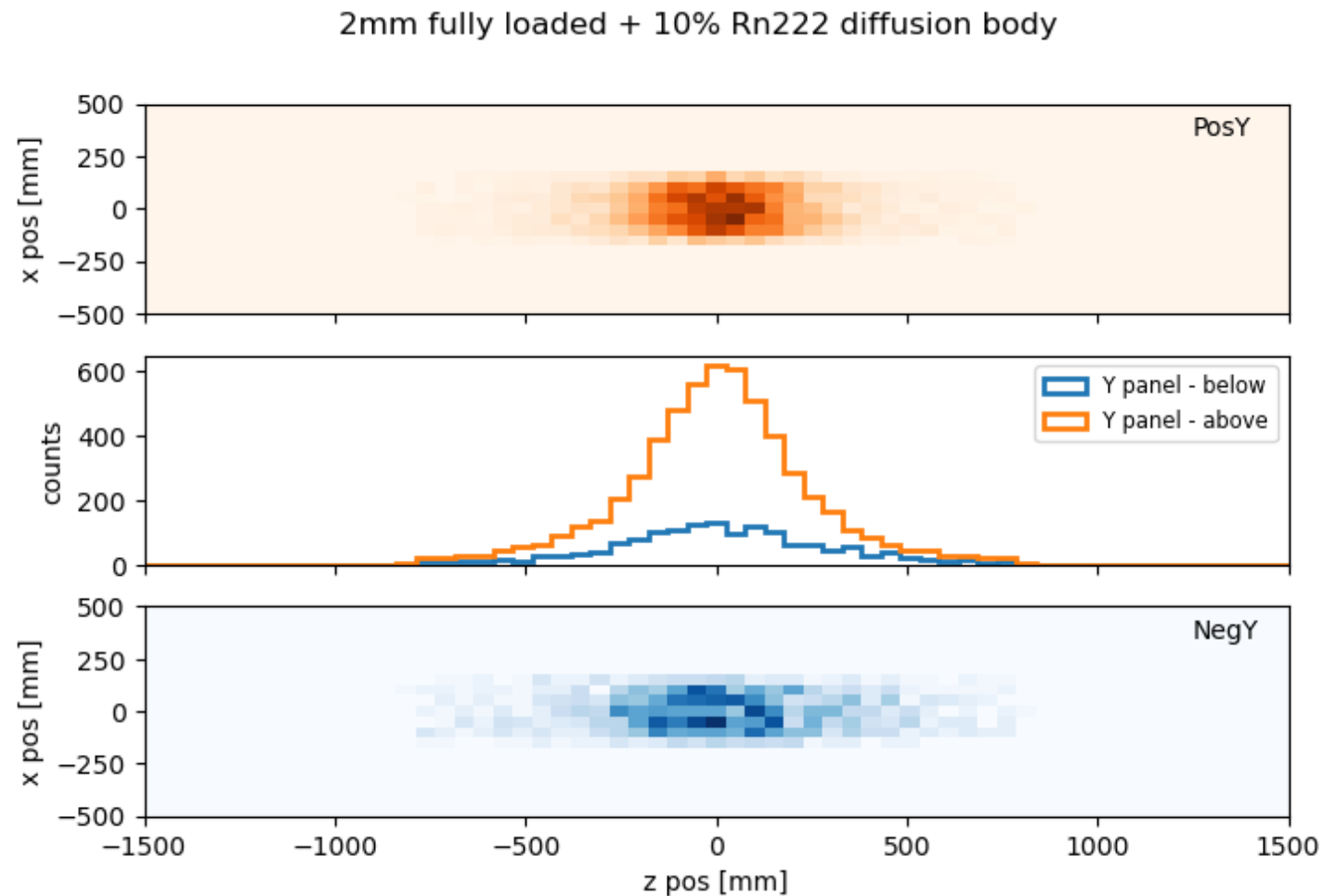
- Visualising different GPSs



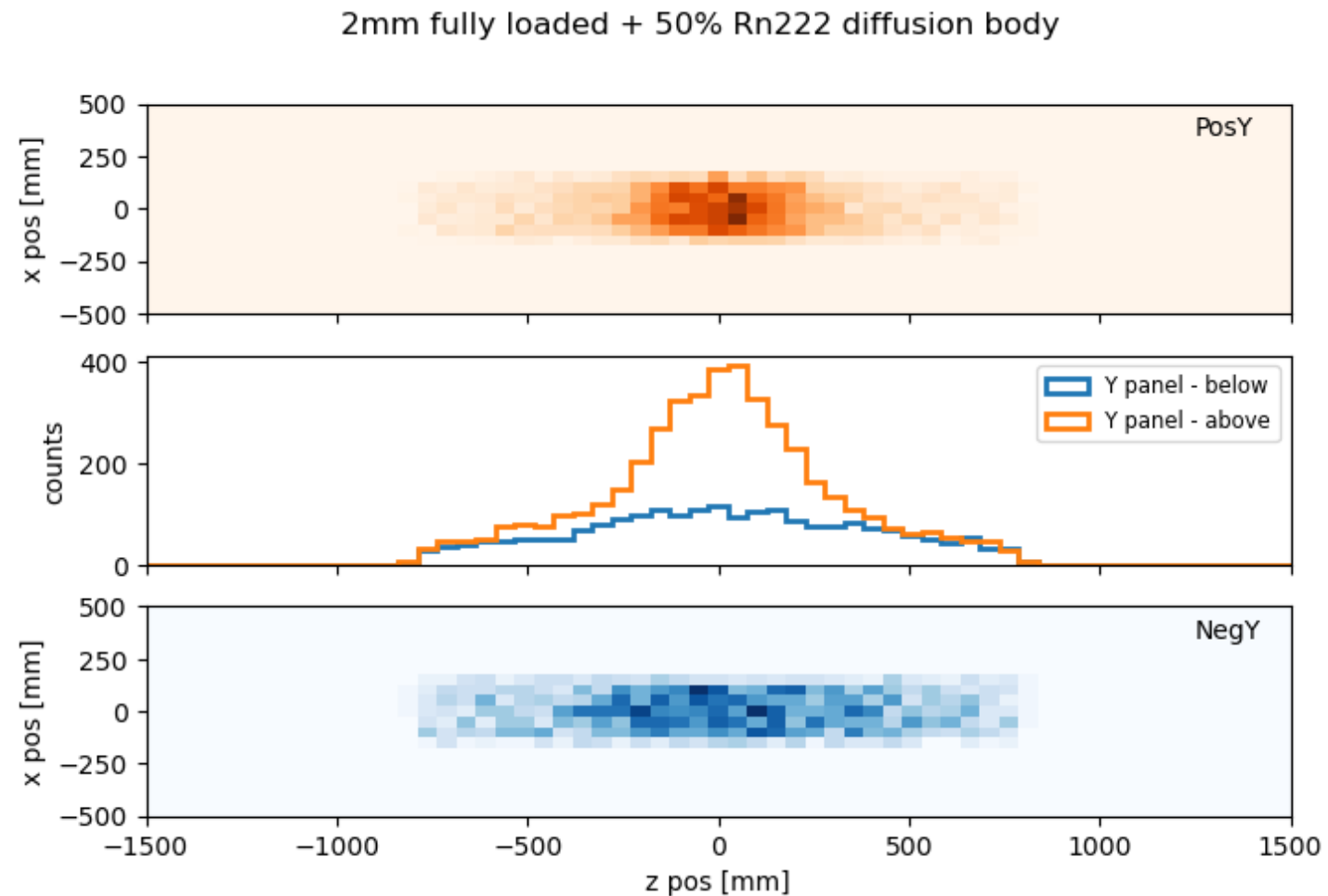
Fully loaded 2mm glue on tissue



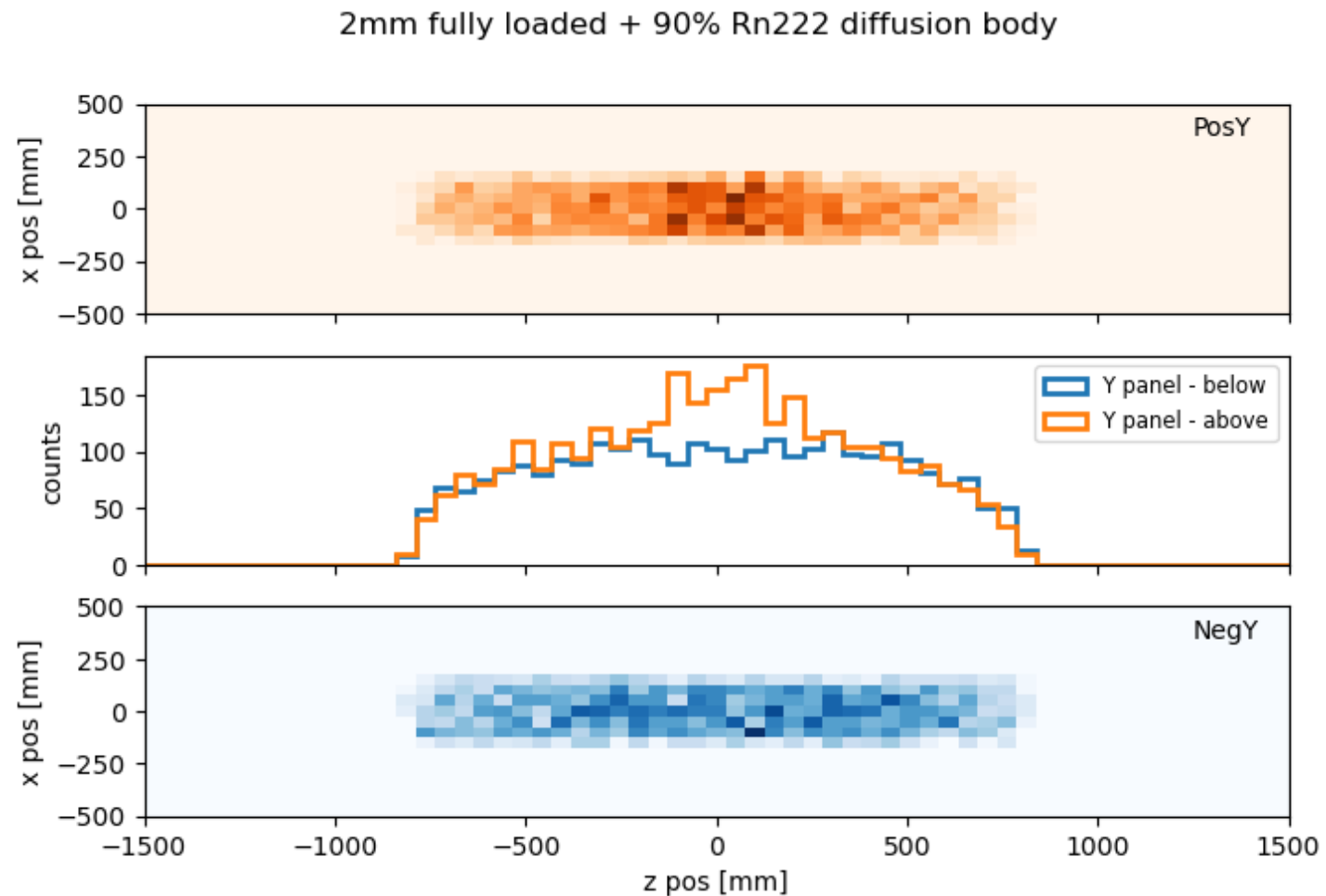
Fully loaded 2mm glue on tissue + 10% Rn222 diffusion



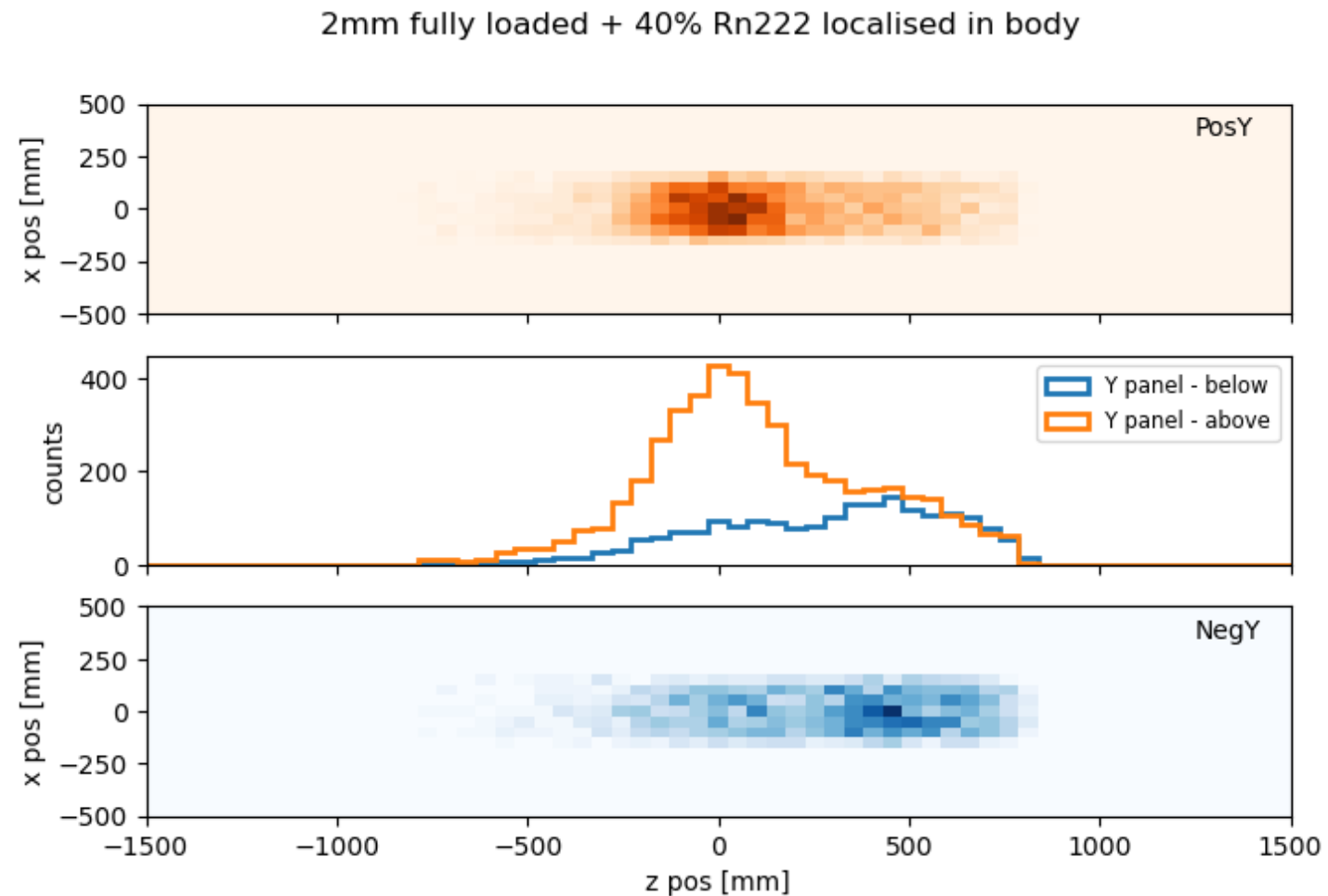
Fully loaded 2mm glue on tissue + 50% Rn222 diffusion



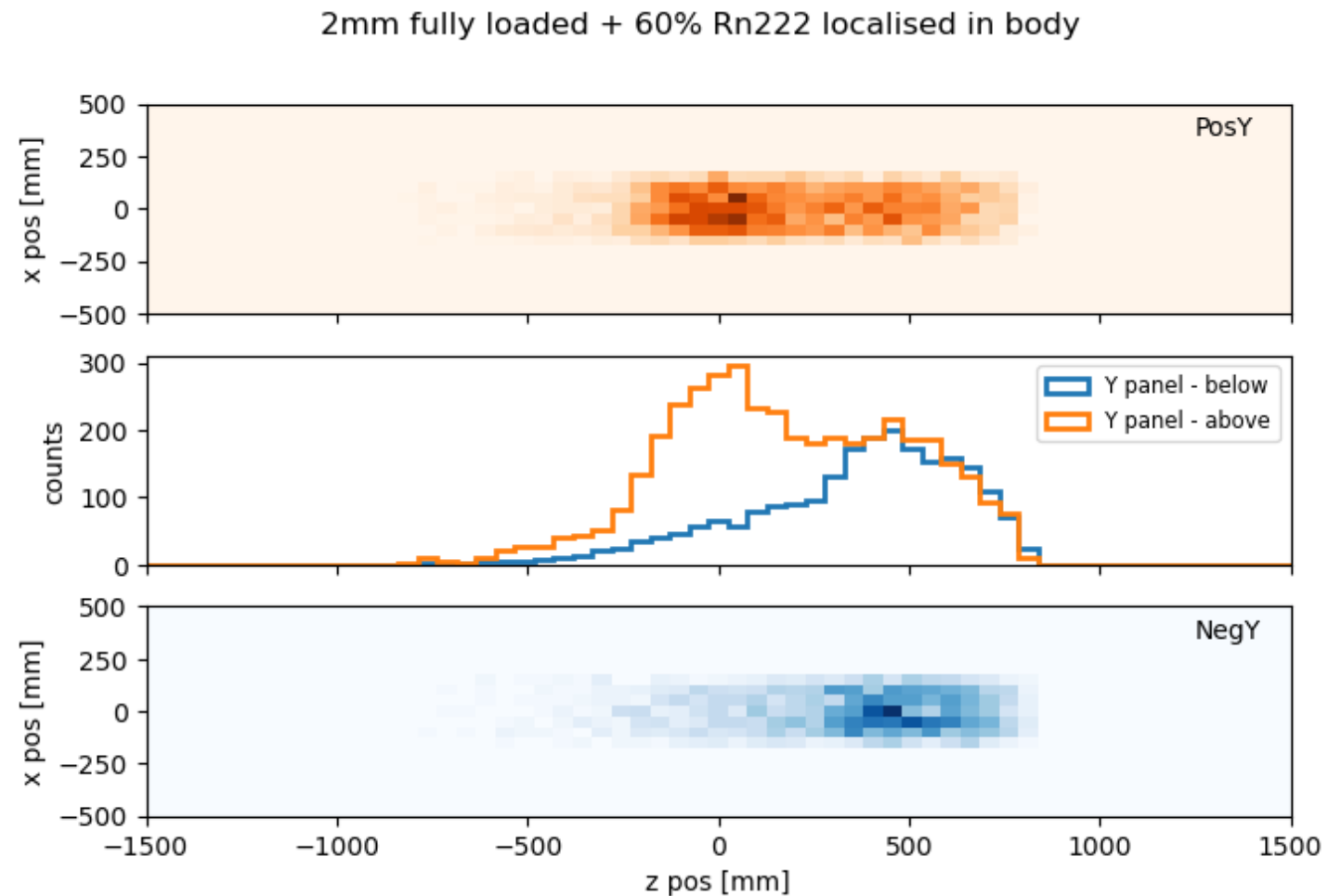
Fully loaded 2mm glue on tissue + 90% Rn222 diffusion



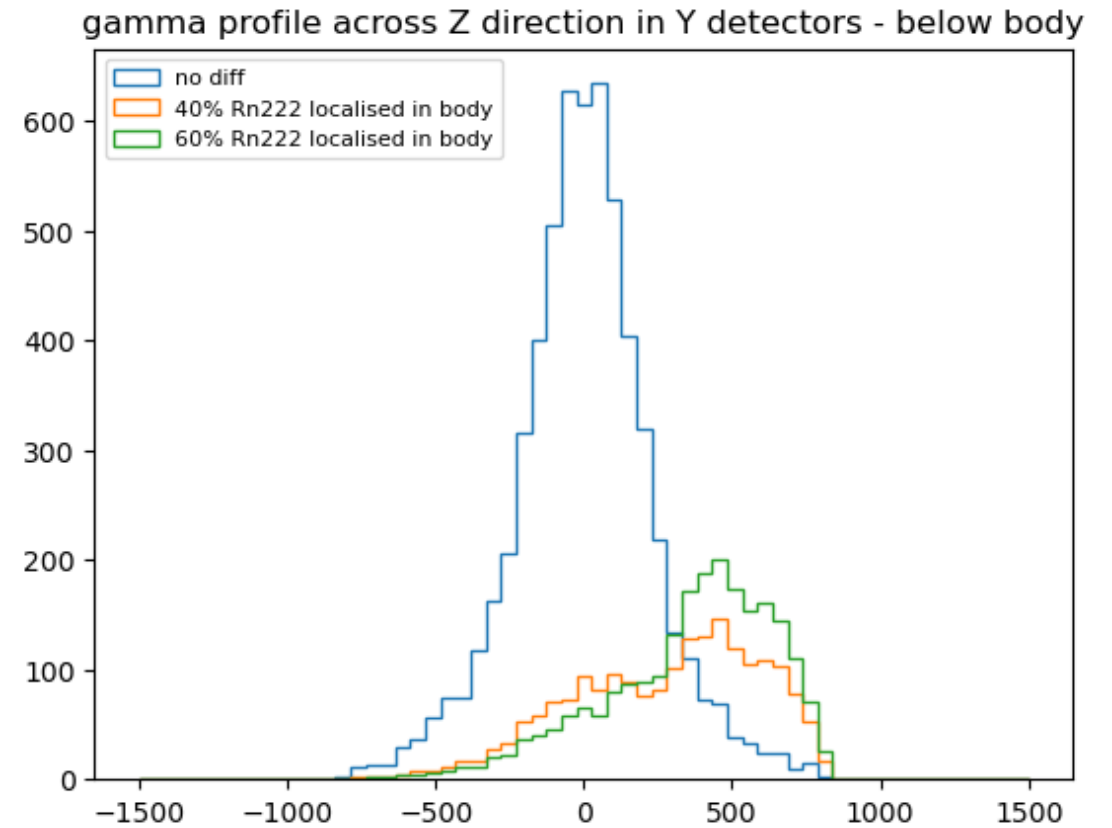
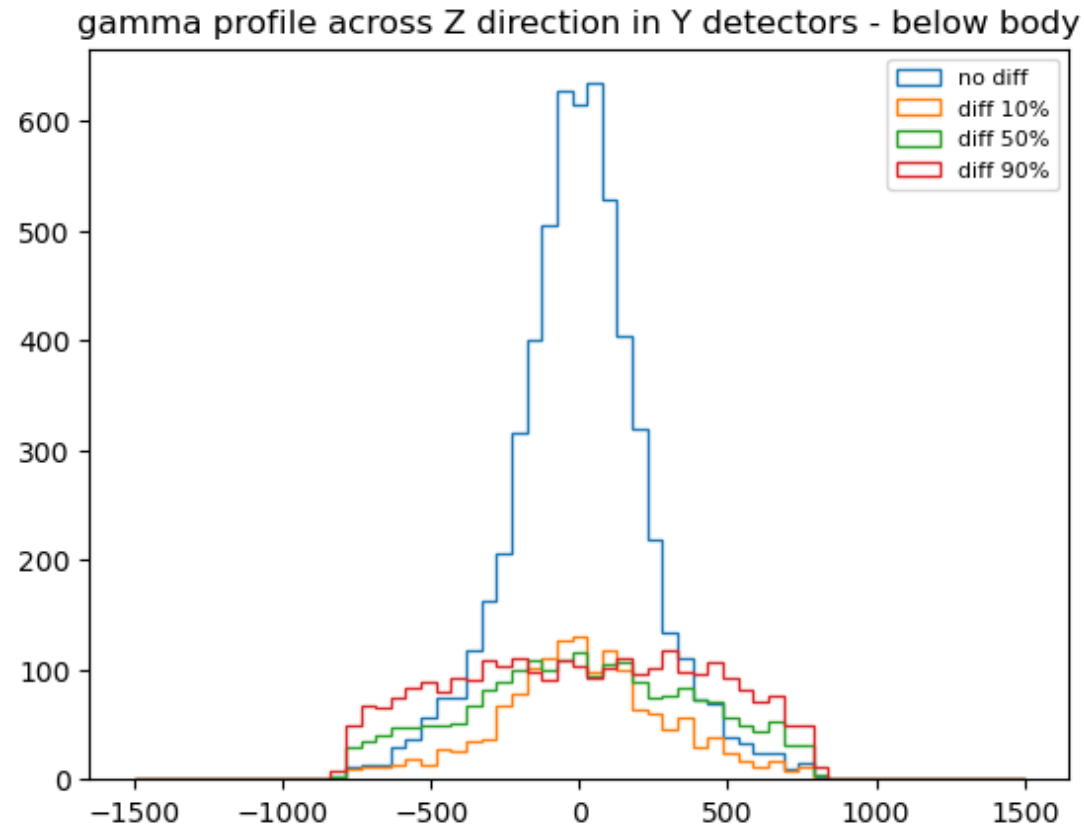
Fully loaded 2mm glue on tissue 40% Rn222 localised in body



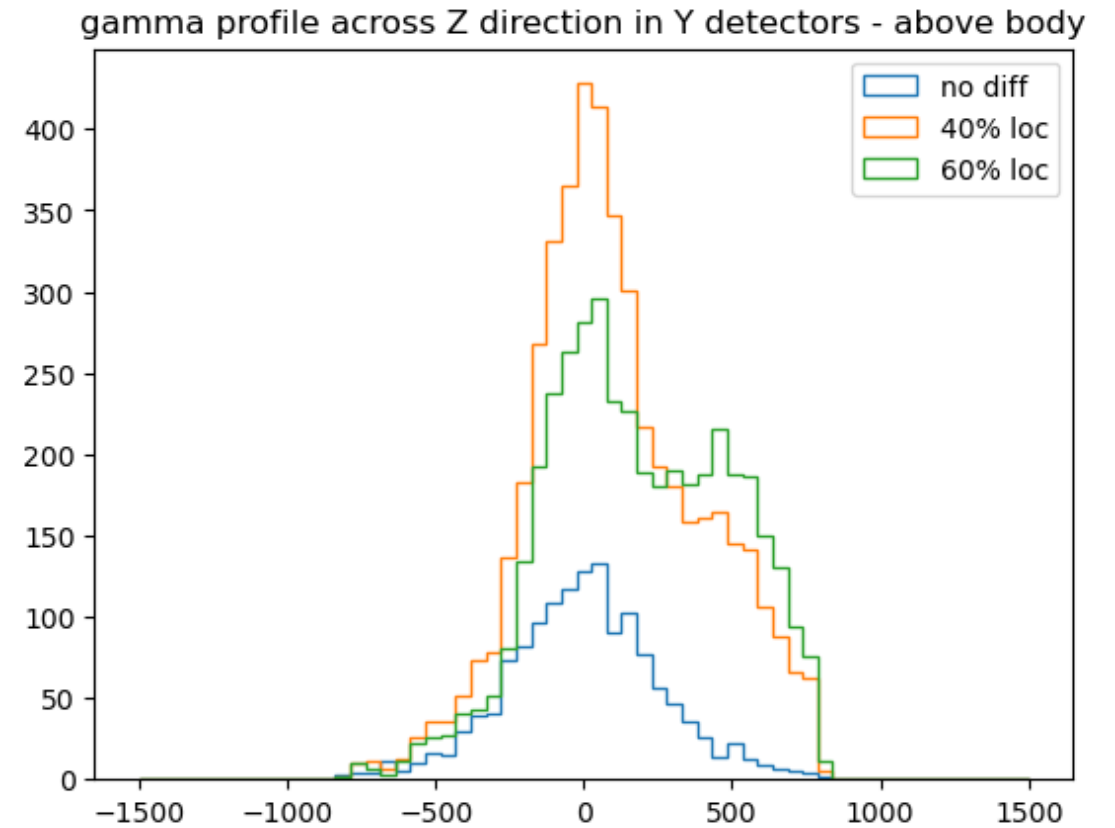
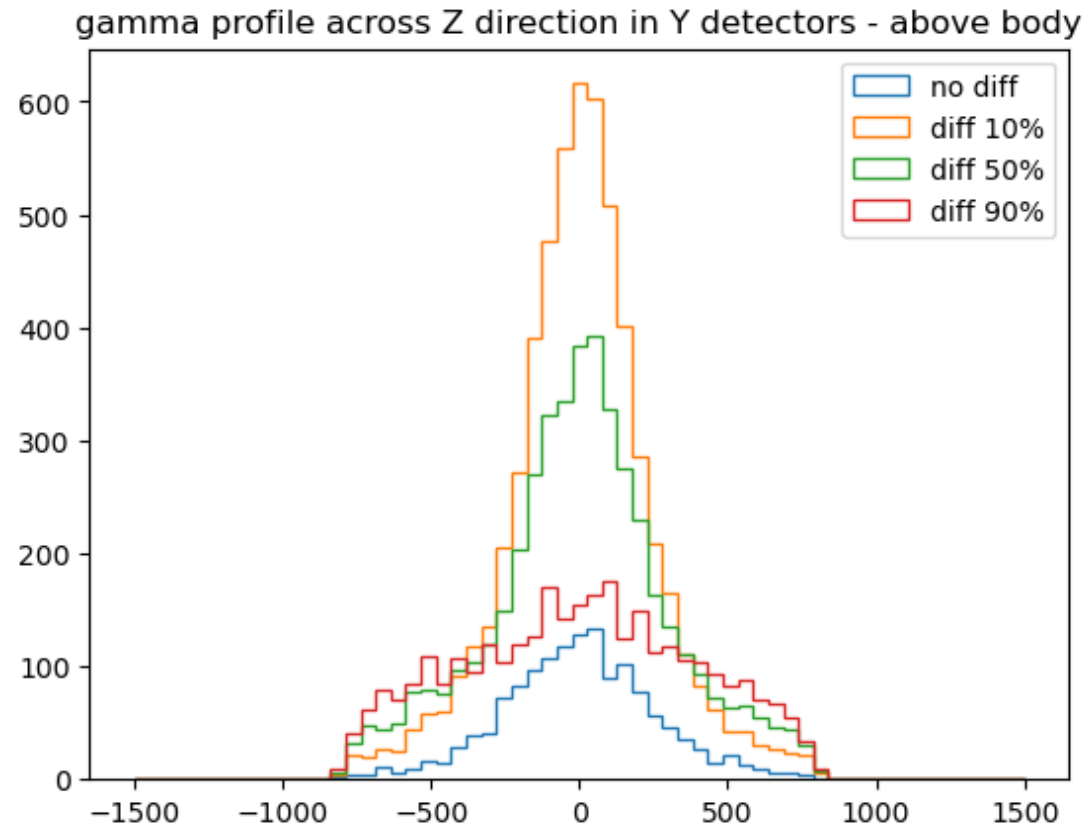
Fully loaded 2mm glue on tissue 60% Rn222 localised in body



Profile comparison – upper detector



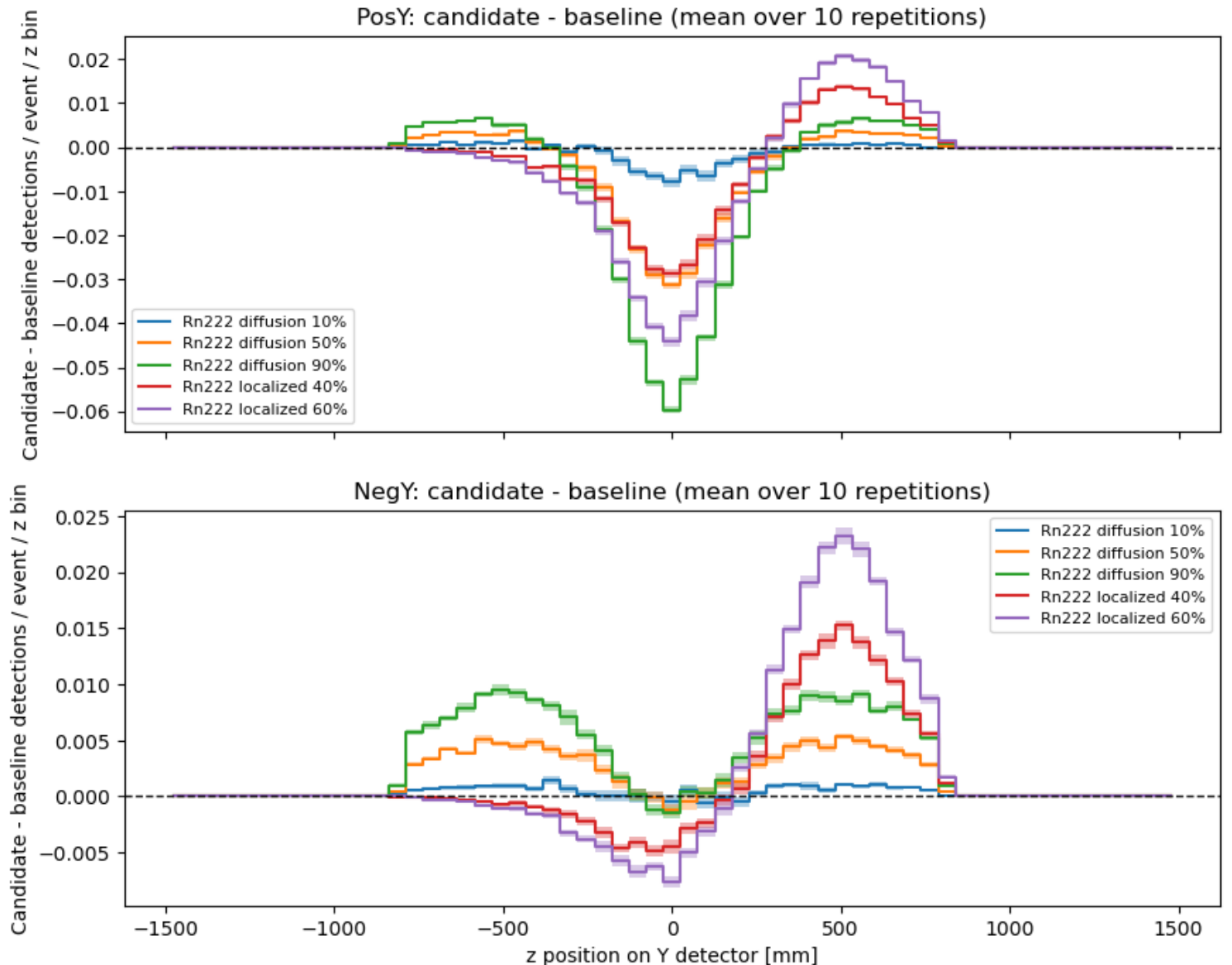
Profile comparison – lower detector



Signal detection

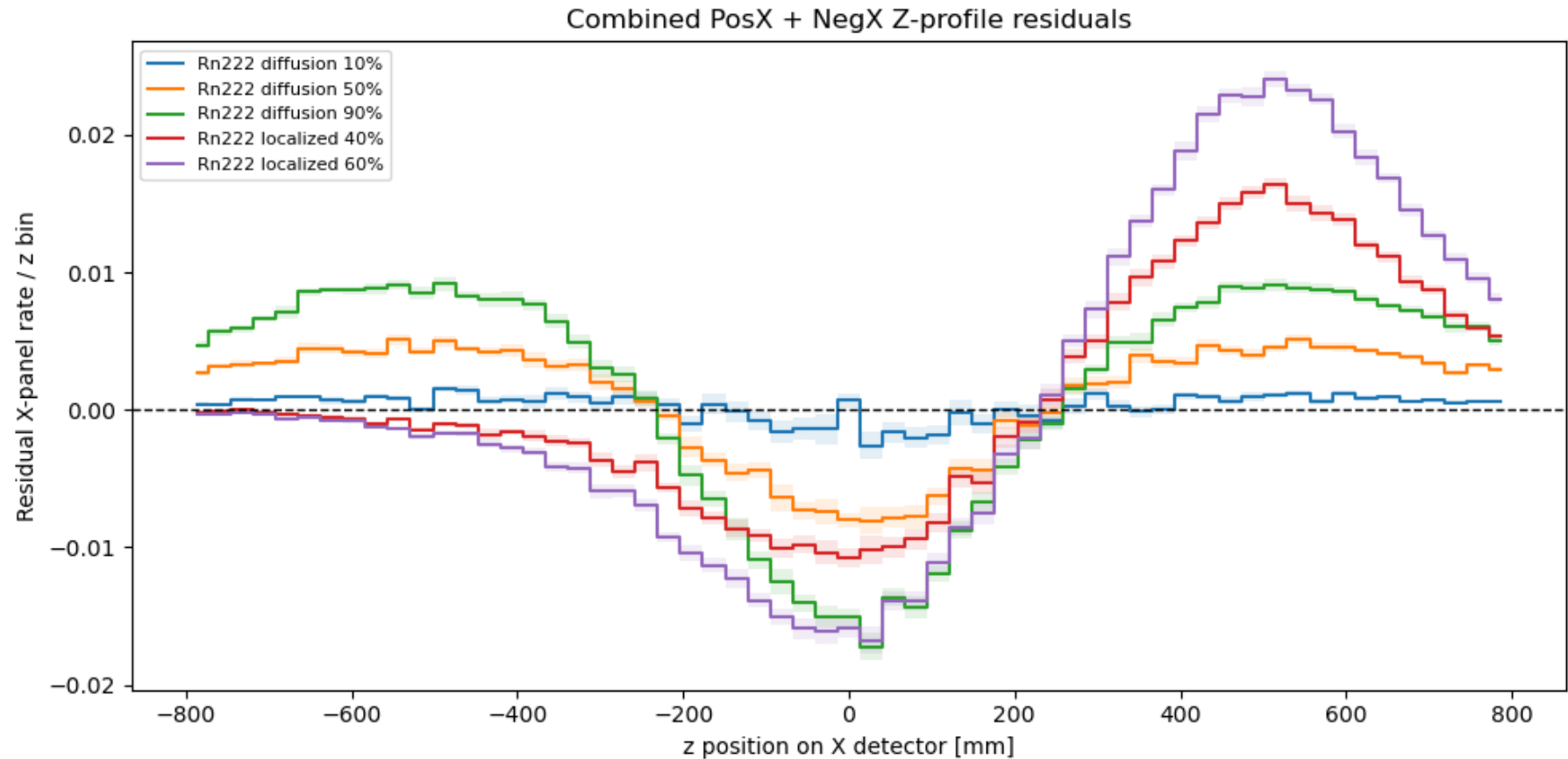
- In each bin:
No. gamma counts/no.
events (10k) for candidate –
baseline
- Baseline: 2mm fully loaded
glue placed on top of the
body, in the centre
- Diffusion lowers number of
counts towards the centre
of the area.
- Localised emitter increases
the number of counts in
localised area

Baseline-subtracted Z profiles, normalization= No. events

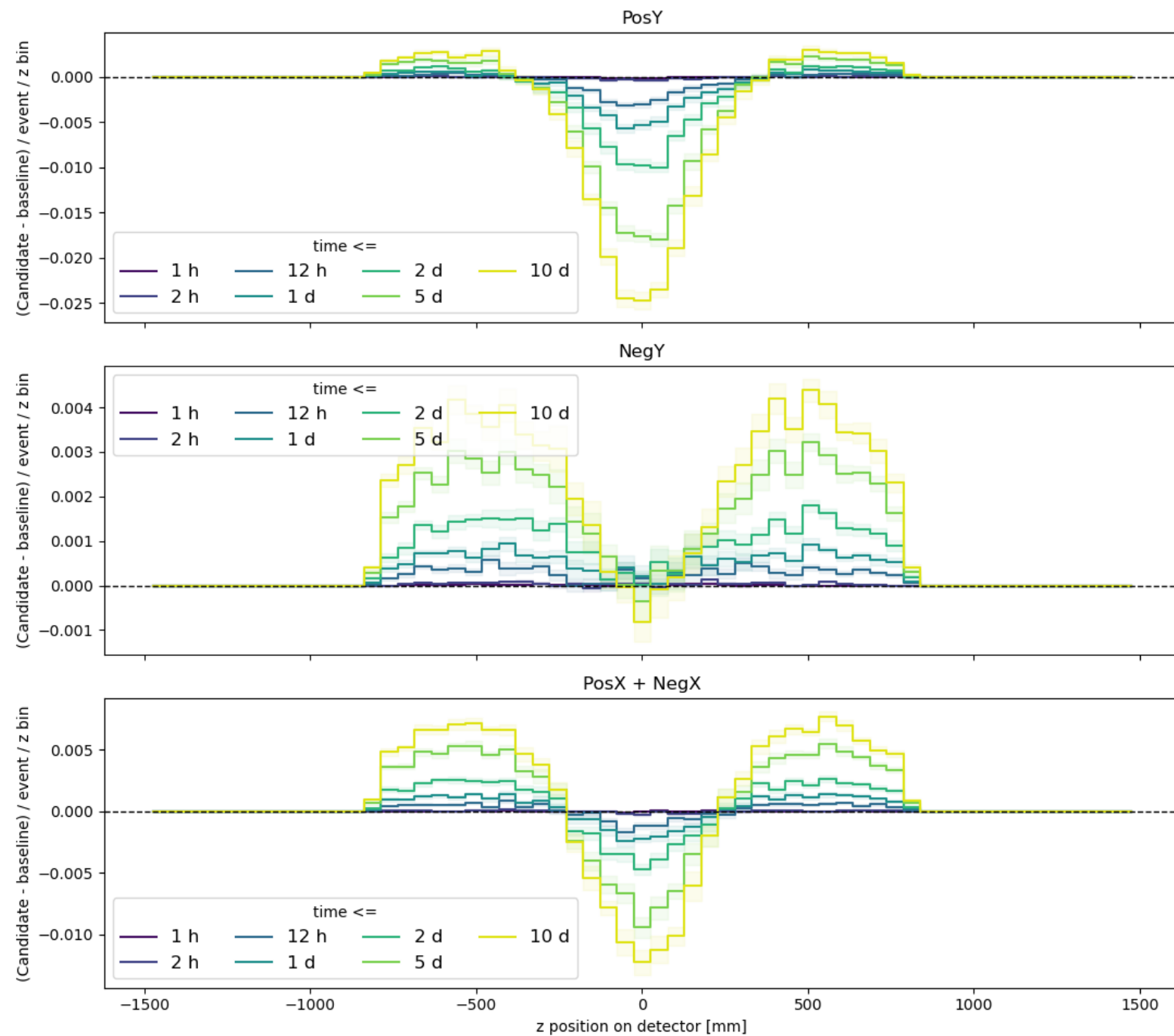


Signal detection

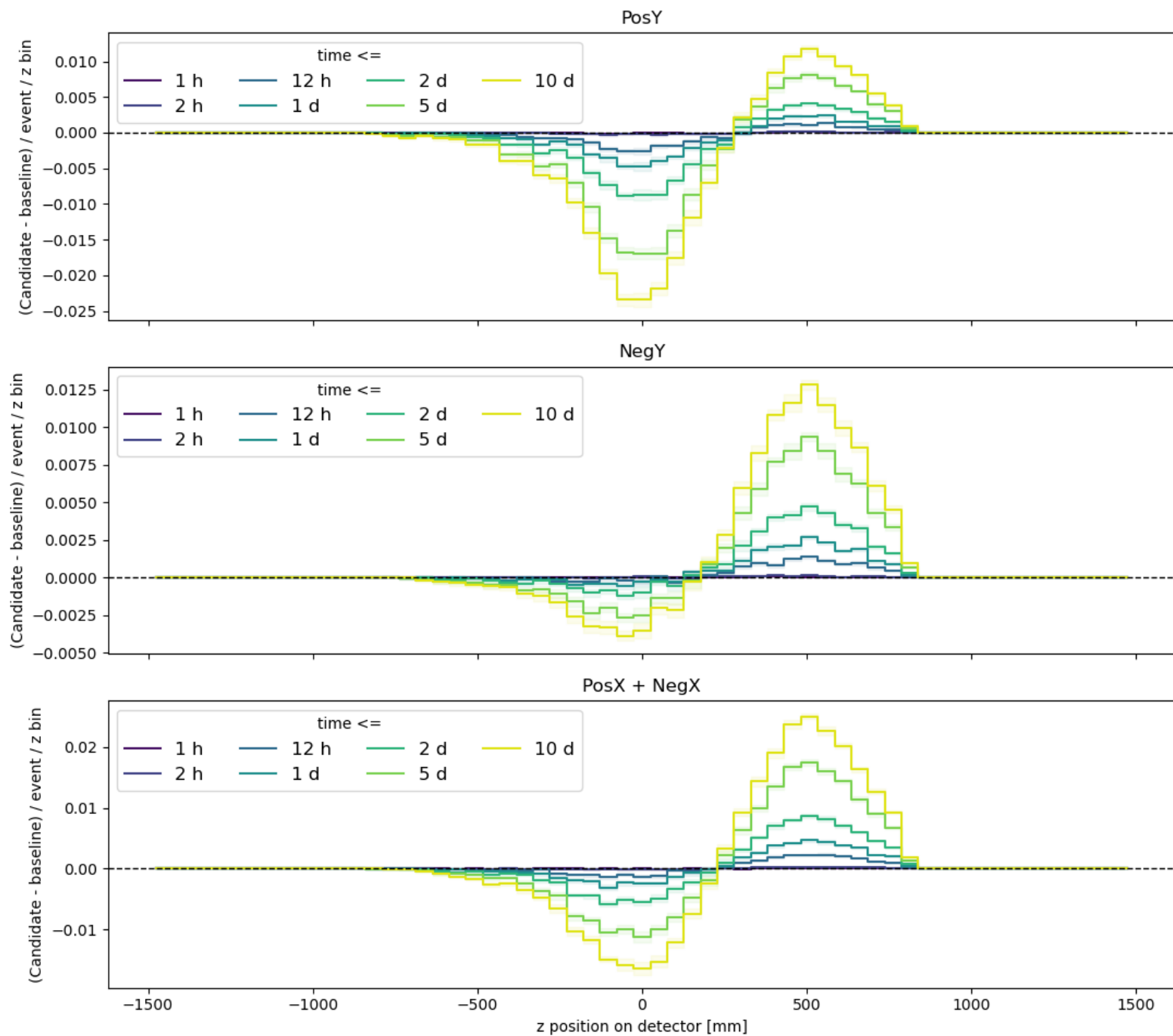
- X panel, Z profile



Cumulative time-window residual Z profiles: Rn222 diffusion 50% vs baseline



Cumulative time-window residual Z profiles: Rn222 localized 40% vs baseline



2D Y-panel residual maps: Rn222 localized 60% vs baseline

